

Summary Tables of MAXCUT Data

1 Tables for depth $P = 1$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp. (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	-	-	-
TQA (Aer)	-	-	-	-
TQA*	-	-	-	-
TQA* (Aer)	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	55 [10–20]	-	-	273 [10–20]
Fixed Angle [†]	55 [10–20]	-	-	289 [10–20]
Fourier*	61 [10–20]	10 [12]	699 [10–20]	570 [10–20]
Interp.	83 [10–20]	10 [12]	805 [10–22]	382 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	81 [10–20]	10 [12]	481 [10–22]	368 [10–20]
TQA	65 [10–20]	-	592 [10–17]	280 [10–20]
TQA*	63 [10–20]	-	584 [10–17]	60 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Interp.	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Interp. (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3
TQA	-	-	-	-
TQA (Aer)	-	-	-	-
TQA*	-	-	-	-
TQA* (Aer)	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Interp.	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	79.1 ± 2.9	-	-	81.7 ± 4.3
Fixed Angle [†]	78.4 ± 2.6	-	-	80.7 ± 3.1
Fourier*	78.8 ± 3.1	79.3 ± 3.0	74.8 ± 4.2	82.0 ± 4.1
Interp.	78.8 ± 2.9	79.3 ± 3.0	74.6 ± 3.9	81.9 ± 4.2
Linear Ramp*	-	-	-	-
Recursive TS	78.8 ± 2.9	79.3 ± 3.0	75.0 ± 4.8	82.0 ± 4.3
TQA	77.6 ± 2.6	-	71.0 ± 5.1	79.0 ± 2.9
TQA*	78.8 ± 3.0	-	74.7 ± 3.6	78.1 ± 2.1

Table 2: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	RR
MPS								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-	-	-	-	-
Fourier (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [39–144]
Interp.	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [39–144]
Interp. (Aer)	81.1 ± 1.2	75.6 ± 1.2	73.7 ± 2.2	76.3 ± 1.4	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp*	-	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3	-	1 [106]	-	2 [40]
TQA	-	-	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
TQA* (Aer)	-	-	-	-	-	-	-	-
PP								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [39–144]
Interp.	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [39–144]
Linear Ramp*	-	-	-	-	-	-	-	-
Param. Transfer	-	-	-	-	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3	-	1 [106]	-	2 [40]
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
SV								
Fixed Angle*	79.1 ± 2.9	-	-	81.7 ± 4.3	55 [10–20]	-	-	273 [10–20]
Fixed Angle [†]	78.4 ± 2.6	-	-	80.7 ± 3.1	55 [10–20]	-	-	289 [10–20]
Fourier*	78.8 ± 3.1	79.3 ± 3.0	74.8 ± 4.2	82.0 ± 4.1	61 [10–20]	10 [12]	699 [10–20]	570 [10–20]
Interp.	78.8 ± 2.9	79.3 ± 3.0	74.6 ± 3.9	81.9 ± 4.2	83 [10–20]	10 [12]	805 [10–22]	382 [10–22]
Linear Ramp*	-	-	-	-	-	-	-	-
Recursive TS	78.8 ± 2.9	79.3 ± 3.0	75.0 ± 4.8	82.0 ± 4.3	81 [10–20]	10 [12]	481 [10–22]	368 [10–22]
TQA	77.6 ± 2.6	-	71.0 ± 5.1	79.0 ± 2.9	65 [10–20]	-	592 [10–17]	280 [10–17]
TQA*	78.8 ± 3.0	-	74.7 ± 3.6	78.1 ± 2.1	63 [10–20]	-	584 [10–17]	60 [10–17]

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

2 Tables for depth $P = 2$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	10 [50]	-	90 [100]	40 [30-70]
Fixed Angle* (Aer)	-	-	20 [100]	10 [40]
Fixed Angle [†]	10 [50]	-	90 [100]	40 [30-70]
Fixed Angle [†] (Aer)	-	-	20 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39-144]	30 [40-100]	10 [40]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	51 [30-100]
Interp.	20 [40-50]	31 [39-144]	130 [40-100]	51 [30-100]
Interp. (Aer)	10 [40]	20 [39-144]	30 [40-100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40-50]	10 [39]	130 [40-100]	50 [30-70]
TQA (Aer)	10 [40]	10 [39]	30 [40-100]	10 [40]
TQA*	20 [40-50]	10 [39]	130 [40-100]	50 [30-70]
TQA* (Aer)	10 [40]	10 [39]	30 [40-100]	10 [40]
PP				
Fixed Angle*	10 [50]	-	81 [100]	40 [30-70]
Fixed Angle [†]	10 [50]	-	81 [100]	40 [30-70]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	51 [30-100]
Interp.	20 [40-50]	31 [39-144]	130 [40-100]	51 [30-100]
Linear Ramp*	-	-	10 [100]	-
Param. Transfer	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40-50]	10 [39]	130 [40-100]	50 [30-70]
TQA*	20 [40-50]	10 [39]	130 [40-100]	50 [30-70]
SV				
Fixed Angle*	72 [10-20]	-	293 [10-22]	360 [10-20]
Fixed Angle [†]	72 [10-20]	-	294 [10-22]	372 [10-20]
Fourier*	61 [10-20]	10 [12]	699 [10-20]	567 [10-20]
Interp.	83 [10-20]	10 [12]	805 [10-22]	372 [10-20]
Linear Ramp*	-	-	-	-
Recursive TS	81 [10-20]	10 [12]	415 [10-22]	353 [10-20]
TQA	83 [10-20]	10 [12]	804 [10-22]	361 [10-20]
TQA*	83 [10-20]	10 [12]	803 [10-22]	136 [10-20]

Table 4: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	80.4 ± 1.8	-	79.9 ± 2.4	77.3 ± 4.4
Fixed Angle* (Aer)	-	-	73.4 ± 1.9	79.0 ± 2.6
Fixed Angle [†]	77.2 ± 0.9	-	79.6 ± 2.6	76.6 ± 4.8
Fixed Angle [†] (Aer)	-	-	66.7 ± 2.4	78.3 ± 2.8
Fourier (Aer)	70.5 ± 2.6	75.6 ± 3.8	69.5 ± 2.0	75.4 ± 2.2
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	75.2 ± 4.4
Interp.	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	75.6 ± 4.0
Interp. (Aer)	77.5 ± 1.8	80.4 ± 2.0	74.4 ± 3.1	71.1 ± 2.5
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	68.4 ± 0.5	-
Recursive TS	-	-	-	-
TQA	69.3 ± 1.1	73.0 ± 1.7	70.0 ± 3.9	72.8 ± 2.5
TQA (Aer)	69.2 ± 2.0	73.1 ± 1.6	67.8 ± 1.3	73.7 ± 1.9
TQA*	77.4 ± 1.3	83.1 ± 1.6	79.2 ± 3.0	77.6 ± 4.0
TQA* (Aer)	75.8 ± 1.3	80.7 ± 2.2	71.5 ± 4.2	77.3 ± 3.4
PP				
Fixed Angle*	87.8 ± 1.0	-	79.2 ± 1.6	83.6 ± 1.4
Fixed Angle [†]	86.9 ± 1.0	-	79.1 ± 1.6	83.6 ± 1.4
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.2 ± 2.5
Interp.	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4
Linear Ramp*	-	-	81.6 ± 1.1	-
Param. Transfer	-	-	82.8 ± 0.8	-
Recursive TS	-	-	-	-
TQA	78.9 ± 1.3	73.0 ± 1.7	70.6 ± 3.0	76.9 ± 1.9
TQA*	87.4 ± 1.2	83.1 ± 1.6	79.6 ± 2.0	83.8 ± 1.4
SV				
Fixed Angle*	85.5 ± 2.3	-	82.6 ± 2.9	88.1 ± 3.3
Fixed Angle [†]	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0
Fourier*	75.8 ± 9.1	77.0 ± 3.9	76.4 ± 5.8	80.6 ± 8.9
Interp.	85.3 ± 2.4	85.5 ± 1.8	82.5 ± 4.1	86.5 ± 4.6
Linear Ramp*	-	-	-	-
Recursive TS	85.2 ± 2.4	87.1 ± 1.9	81.3 ± 2.8	86.1 ± 4.3
TQA	76.0 ± 3.2	73.0 ± 5.0	72.4 ± 4.2	75.7 ± 4.4
TQA*	85.5 ± 2.4	86.6 ± 2.5	82.8 ± 3.1	87.3 ± 3.3

Table 5: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	RR
MPS								
Fixed Angle*	80.4 ± 1.8	-	79.9 ± 2.4	77.3 ± 4.4	10 [50]	-	90 [100]	40 [30]
Fixed Angle* (Aer)	-	-	73.4 ± 1.9	79.0 ± 2.6	-	-	20 [100]	10 [30]
Fixed Angle†	77.2 ± 0.9	-	79.6 ± 2.6	76.6 ± 4.8	10 [50]	-	90 [100]	40 [30]
Fixed Angle† (Aer)	-	-	66.7 ± 2.4	78.3 ± 2.8	-	-	20 [100]	10 [30]
Fourier (Aer)	70.5 ± 2.6	75.6 ± 3.8	69.5 ± 2.0	75.4 ± 2.2	10 [40]	20 [39–144]	30 [40–100]	10 [30]
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	75.2 ± 4.4	20 [40–50]	31 [39–144]	130 [40–100]	51 [30]
Interp.	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	75.6 ± 4.0	20 [40–50]	31 [39–144]	130 [40–100]	51 [30]
Interp. (Aer)	77.5 ± 1.8	80.4 ± 2.0	74.4 ± 3.1	71.1 ± 2.5	10 [40]	20 [39–144]	30 [40–100]	10 [30]
Linear Ramp*	-	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	68.4 ± 0.5	-	-	-	10 [100]	-
Recursive TS	-	-	-	-	-	-	-	-
TQA	69.3 ± 1.1	73.0 ± 1.7	70.0 ± 3.9	72.8 ± 2.5	20 [40–50]	10 [39]	130 [40–100]	50 [30]
TQA (Aer)	69.2 ± 2.0	73.1 ± 1.6	67.8 ± 1.3	73.7 ± 1.9	10 [40]	10 [39]	30 [40–100]	10 [30]
TQA*	77.4 ± 1.3	83.1 ± 1.6	79.2 ± 3.0	77.6 ± 4.0	20 [40–50]	10 [39]	130 [40–100]	50 [30]
TQA* (Aer)	75.8 ± 1.3	80.7 ± 2.2	71.5 ± 4.2	77.3 ± 3.4	10 [40]	10 [39]	30 [40–100]	10 [30]
PP								
Fixed Angle*	87.8 ± 1.0	-	79.2 ± 1.6	83.6 ± 1.4	10 [50]	-	81 [100]	40 [30]
Fixed Angle†	86.9 ± 1.0	-	79.1 ± 1.6	83.6 ± 1.4	10 [50]	-	81 [100]	40 [30]
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.2 ± 2.5	20 [40–50]	31 [39–144]	130 [40–100]	51 [30]
Interp.	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	51 [30]
Linear Ramp*	-	-	81.6 ± 1.1	-	-	-	10 [100]	-
Param. Transfer	-	-	82.8 ± 0.8	-	-	-	10 [100]	-
Recursive TS	-	-	-	-	-	-	-	-
TQA	78.9 ± 1.3	73.0 ± 1.7	70.6 ± 3.0	76.9 ± 1.9	20 [40–50]	10 [39]	130 [40–100]	50 [30]
TQA*	87.4 ± 1.2	83.1 ± 1.6	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	10 [39]	130 [40–100]	50 [30]
SV								
Fixed Angle*	85.5 ± 2.3	-	82.6 ± 2.9	88.1 ± 3.3	72 [10–20]	-	293 [10–22]	360 [10–22]
Fixed Angle†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0	72 [10–20]	-	294 [10–22]	372 [10–22]
Fourier*	75.8 ± 9.1	77.0 ± 3.9	76.4 ± 5.8	80.6 ± 8.9	61 [10–20]	10 [12]	699 [10–20]	567 [10–20]
Interp.	85.3 ± 2.4	85.5 ± 1.8	82.5 ± 4.1	86.5 ± 4.6	83 [10–20]	10 [12]	805 [10–22]	372 [10–22]
Linear Ramp*	-	-	-	-	-	-	-	-
Recursive TS	85.2 ± 2.4	87.1 ± 1.9	81.3 ± 2.8	86.1 ± 4.3	81 [10–20]	10 [12]	415 [10–22]	353 [10–22]
TQA	76.0 ± 3.2	73.0 ± 5.0	72.4 ± 4.2	75.7 ± 4.4	83 [10–20]	10 [12]	804 [10–22]	361 [10–22]
TQA*	85.5 ± 2.4	86.6 ± 2.5	82.8 ± 3.1	87.3 ± 3.3	83 [10–20]	10 [12]	803 [10–22]	136 [10–22]

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

3 Tables for depth $P = 3$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	30 [30–70]
Fixed Angle* (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	30 [30–70]
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Interp. (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	10 [40]	30 [30–70]
TQA (Aer)	-	-	-	-
TQA*	-	-	10 [40]	30 [30–70]
TQA* (Aer)	-	-	-	-
PP				
Fixed Angle*	-	-	-	30 [30–70]
Fixed Angle [†]	-	-	-	30 [30–70]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	10 [40]	30 [30–70]
TQA*	-	-	10 [40]	30 [30–70]
SV				
Fixed Angle*	72 [10–20]	-	266 [10–20]	364 [10–20]
Fixed Angle [†]	72 [10–20]	-	266 [10–20]	372 [10–20]
Fourier*	59 [10–19]	10 [12]	699 [10–20]	567 [10–20]
Interp.	82 [10–20]	10 [12]	803 [10–22]	372 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	81 [10–20]	10 [12]	415 [10–22]	357 [10–20]
TQA	83 [10–20]	10 [12]	803 [10–22]	357 [10–20]
TQA*	82 [10–20]	10 [12]	803 [10–22]	135 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	83.3 ± 5.2
Fixed Angle* (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	81.6 ± 6.2
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	80.0 ± 4.1	77.9 ± 2.7	71.6 ± 6.0	77.1 ± 2.5
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	78.0 ± 5.3
Interp.	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	81.4 ± 6.7
Interp. (Aer)	83.4 ± 3.0	83.4 ± 3.0	82.3 ± 4.8	73.4 ± 3.8
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	82.1 ± 1.8	76.2 ± 3.9
TQA (Aer)	-	-	-	-
TQA*	-	-	88.8 ± 1.1	81.5 ± 6.3
TQA* (Aer)	-	-	-	-
PP				
Fixed Angle*	-	-	-	87.3 ± 1.3
Fixed Angle [†]	-	-	-	87.3 ± 1.3
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.6 ± 2.3
Interp.	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.3 ± 1.5
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	68.0 ± 1.6	80.7 ± 1.4
TQA*	-	-	79.7 ± 2.4	87.3 ± 1.3
SV				
Fixed Angle*	89.1 ± 1.9	-	87.8 ± 2.5	91.6 ± 2.7
Fixed Angle [†]	87.1 ± 1.8	-	87.2 ± 2.5	89.9 ± 2.2
Fourier*	83.2 ± 5.9	79.2 ± 4.5	78.9 ± 7.6	88.4 ± 4.1
Interp.	88.8 ± 2.0	89.8 ± 1.5	87.4 ± 3.8	89.7 ± 4.1
Linear Ramp*	-	-	-	-
Recursive TS	88.0 ± 2.5	90.1 ± 1.8	84.9 ± 2.4	88.5 ± 3.8
TQA	81.1 ± 4.0	79.7 ± 4.8	77.3 ± 5.2	80.1 ± 5.5
TQA*	87.6 ± 2.2	88.2 ± 2.0	85.1 ± 3.8	89.9 ± 2.6

Table 8: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	RR
MPS								
Fixed Angle*	-	-	-	83.3 ± 5.2	-	-	-	30 [3]
Fixed Angle* (Aer)	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	81.6 ± 6.2	-	-	-	30 [3]
Fixed Angle [†] (Aer)	-	-	-	-	-	-	-	-
Fourier (Aer)	80.0 ± 4.1	77.9 ± 2.7	71.6 ± 6.0	77.1 ± 2.5	10 [40]	20 [39–144]	30 [40–100]	10 [3]
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	78.0 ± 5.3	20 [40–50]	31 [39–144]	130 [40–100]	41 [3]
Interp.	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	81.4 ± 6.7	20 [40–50]	31 [39–144]	130 [40–100]	41 [3]
Interp. (Aer)	83.4 ± 3.0	83.4 ± 3.0	82.3 ± 4.8	73.4 ± 3.8	10 [40]	20 [39–144]	30 [40–100]	10 [3]
Linear Ramp*	-	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-	-
TQA	-	-	82.1 ± 1.8	76.2 ± 3.9	-	-	10 [40]	30 [3]
TQA (Aer)	-	-	-	-	-	-	-	-
TQA*	-	-	88.8 ± 1.1	81.5 ± 6.3	-	-	10 [40]	30 [3]
TQA* (Aer)	-	-	-	-	-	-	-	-
PP								
Fixed Angle*	-	-	-	87.3 ± 1.3	-	-	-	30 [3]
Fixed Angle [†]	-	-	-	87.3 ± 1.3	-	-	-	30 [3]
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.6 ± 2.3	20 [40–50]	31 [39–144]	130 [40–100]	41 [3]
Interp.	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.3 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	41 [3]
Linear Ramp*	-	-	-	-	-	-	-	-
Param. Transfer	-	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-	-
TQA	-	-	68.0 ± 1.6	80.7 ± 1.4	-	-	10 [40]	30 [3]
TQA*	-	-	79.7 ± 2.4	87.3 ± 1.3	-	-	10 [40]	30 [3]
SV								
Fixed Angle*	89.1 ± 1.9	-	87.8 ± 2.5	91.6 ± 2.7	72 [10–20]	-	266 [10–20]	364 [3]
Fixed Angle [†]	87.1 ± 1.8	-	87.2 ± 2.5	89.9 ± 2.2	72 [10–20]	-	266 [10–20]	372 [3]
Fourier*	83.2 ± 5.9	79.2 ± 4.5	78.9 ± 7.6	88.4 ± 4.1	59 [10–19]	10 [12]	699 [10–20]	567 [3]
Interp.	88.8 ± 2.0	89.8 ± 1.5	87.4 ± 3.8	89.7 ± 4.1	82 [10–20]	10 [12]	803 [10–22]	372 [3]
Linear Ramp*	-	-	-	-	-	-	-	-
Recursive TS	88.0 ± 2.5	90.1 ± 1.8	84.9 ± 2.4	88.5 ± 3.8	81 [10–20]	10 [12]	415 [10–22]	357 [3]
TQA	81.1 ± 4.0	79.7 ± 4.8	77.3 ± 5.2	80.1 ± 5.5	83 [10–20]	10 [12]	803 [10–22]	357 [3]
TQA*	87.6 ± 2.2	88.2 ± 2.0	85.1 ± 3.8	89.9 ± 2.6	82 [10–20]	10 [12]	803 [10–22]	135 [3]

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

4 Tables for depth $P = 4$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	30 [100]	10 [40]
Fixed Angle* (Aer)	-	-	20 [100]	10 [40]
Fixed Angle [†]	-	-	30 [100]	10 [40]
Fixed Angle [†] (Aer)	-	-	20 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA*	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA* (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
PP				
Fixed Angle*	-	-	20 [100]	10 [40]
Fixed Angle [†]	-	-	20 [100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	130 [40–100]	20 [40]
SV				
Fixed Angle*	52 [10–20]	-	95 [10–20]	155 [10–20]
Fixed Angle [†]	52 [10–20]	-	95 [10–20]	158 [10–20]
Fourier*	59 [10–19]	10 [12]	699 [10–20]	567 [10–20]
Interp.	82 [10–20]	10 [12]	803 [10–22]	372 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	81 [10–20]	10 [12]	414 [10–22]	352 [10–20]
TQA	83 [10–20]	10 [12]	803 [10–22]	354 [10–20]
TQA*	82 [10–20]	10 [12]	803 [10–22]	136 [10–20]

Table 10: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	90.5 ± 1.4	92.6 ± 0.8
Fixed Angle* (Aer)	-	-	81.7 ± 6.6	93.1 ± 1.0
Fixed Angle [†]	-	-	90.4 ± 1.5	90.4 ± 1.2
Fixed Angle [†] (Aer)	-	-	74.8 ± 6.2	91.2 ± 1.7
Fourier (Aer)	74.3 ± 4.1	80.9 ± 2.7	76.9 ± 7.6	79.7 ± 3.3
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	88.0 ± 6.0
Interp.	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	90.5 ± 6.0
Interp. (Aer)	87.8 ± 2.2	85.7 ± 3.4	88.1 ± 4.3	80.4 ± 6.6
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	70.5 ± 3.4	82.0 ± 1.6	84.2 ± 2.4	80.0 ± 1.8
TQA (Aer)	68.9 ± 1.9	81.5 ± 1.0	74.7 ± 8.2	78.5 ± 2.8
TQA*	79.0 ± 7.0	86.9 ± 1.1	88.5 ± 1.9	88.1 ± 3.4
TQA* (Aer)	80.7 ± 4.4	84.8 ± 0.9	77.9 ± 9.6	82.5 ± 3.4
PP				
Fixed Angle*	-	-	86.1 ± 1.7	89.9 ± 1.6
Fixed Angle [†]	-	-	84.1 ± 1.4	89.9 ± 1.6
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.4 ± 1.9
Interp.	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.3 ± 1.6
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	83.7 ± 1.4	82.0 ± 1.6	75.7 ± 4.8	85.4 ± 1.8
TQA*	89.2 ± 1.9	87.3 ± 1.3	83.5 ± 1.9	88.0 ± 1.5
SV				
Fixed Angle*	91.5 ± 1.7	-	91.9 ± 2.3	93.4 ± 2.2
Fixed Angle [†]	89.0 ± 2.2	-	91.0 ± 2.3	92.8 ± 2.2
Fourier*	78.3 ± 10.6	83.6 ± 3.3	80.9 ± 8.0	90.3 ± 4.4
Interp.	91.3 ± 1.7	91.8 ± 1.5	90.4 ± 3.8	91.0 ± 4.8
Linear Ramp*	-	-	-	-
Recursive TS	90.3 ± 2.2	91.4 ± 1.7	87.6 ± 2.3	90.1 ± 3.5
TQA	84.6 ± 3.9	82.0 ± 4.9	79.8 ± 6.0	84.0 ± 5.6
TQA*	89.4 ± 1.9	89.5 ± 1.9	87.8 ± 2.7	91.1 ± 2.5

Table 11: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
MPS								
Fixed Angle*	-	-	90.5 ± 1.4	92.6 ± 0.8	-	-	30 [100]	10
Fixed Angle* (Aer)	-	-	81.7 ± 6.6	93.1 ± 1.0	-	-	20 [100]	10
Fixed Angle†	-	-	90.4 ± 1.5	90.4 ± 1.2	-	-	30 [100]	10
Fixed Angle† (Aer)	-	-	74.8 ± 6.2	91.2 ± 1.7	-	-	20 [100]	10
Fourier (Aer)	74.3 ± 4.1	80.9 ± 2.7	76.9 ± 7.6	79.7 ± 3.3	10 [40]	20 [39–144]	30 [40–100]	10
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	88.0 ± 6.0	20 [40–50]	31 [39–144]	130 [40–100]	21 [100]
Interp.	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	90.5 ± 6.0	20 [40–50]	31 [39–144]	130 [40–100]	21 [100]
Interp. (Aer)	87.8 ± 2.2	85.7 ± 3.4	88.1 ± 4.3	80.4 ± 6.6	10 [40]	20 [39–144]	30 [40–100]	10
Linear Ramp*	-	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-	-
TQA	70.5 ± 3.4	82.0 ± 1.6	84.2 ± 2.4	80.0 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	20 [100]
TQA (Aer)	68.9 ± 1.9	81.5 ± 1.0	74.7 ± 8.2	78.5 ± 2.8	10 [40]	10 [39]	30 [40–100]	10
TQA*	79.0 ± 7.0	86.9 ± 1.1	88.5 ± 1.9	88.1 ± 3.4	20 [40–50]	10 [39]	130 [40–100]	20 [100]
TQA* (Aer)	80.7 ± 4.4	84.8 ± 0.9	77.9 ± 9.6	82.5 ± 3.4	10 [40]	10 [39]	30 [40–100]	10
PP								
Fixed Angle*	-	-	86.1 ± 1.7	89.9 ± 1.6	-	-	20 [100]	10
Fixed Angle†	-	-	84.1 ± 1.4	89.9 ± 1.6	-	-	20 [100]	10
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.4 ± 1.9	20 [40–50]	31 [39–144]	130 [40–100]	21 [100]
Interp.	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.3 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	21 [100]
Linear Ramp*	-	-	-	-	-	-	-	-
Param. Transfer	-	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-	-
TQA	83.7 ± 1.4	82.0 ± 1.6	75.7 ± 4.8	85.4 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	20 [100]
TQA*	89.2 ± 1.9	87.3 ± 1.3	83.5 ± 1.9	88.0 ± 1.5	20 [40–50]	10 [39]	130 [40–100]	20 [100]
SV								
Fixed Angle*	91.5 ± 1.7	-	91.9 ± 2.3	93.4 ± 2.2	52 [10–20]	-	95 [10–20]	155
Fixed Angle†	89.0 ± 2.2	-	91.0 ± 2.3	92.8 ± 2.2	52 [10–20]	-	95 [10–20]	158
Fourier*	78.3 ± 10.6	83.6 ± 3.3	80.9 ± 8.0	90.3 ± 4.4	59 [10–19]	10 [12]	699 [10–20]	567
Interp.	91.3 ± 1.7	91.8 ± 1.5	90.4 ± 3.8	91.0 ± 4.8	82 [10–20]	10 [12]	803 [10–22]	372
Linear Ramp*	-	-	-	-	-	-	-	-
Recursive TS	90.3 ± 2.2	91.4 ± 1.7	87.6 ± 2.3	90.1 ± 3.5	81 [10–20]	10 [12]	414 [10–22]	352
TQA	84.6 ± 3.9	82.0 ± 4.9	79.8 ± 6.0	84.0 ± 5.6	83 [10–20]	10 [12]	803 [10–22]	354
TQA*	89.4 ± 1.9	89.5 ± 1.9	87.8 ± 2.7	91.1 ± 2.5	82 [10–20]	10 [12]	803 [10–22]	136

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

5 Tables for depth $P = 5$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	10 [40]
Fixed Angle* (Aer)	-	10 [144]	-	-
Fixed Angle [†]	-	-	-	10 [40]
Fixed Angle [†] (Aer)	-	10 [144]	-	-
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	10 [144]	10 [100]	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	20 [105–144]	10 [40]	10 [40]
TQA (Aer)	-	10 [144]	-	-
TQA*	-	20 [105–144]	10 [40]	10 [40]
TQA* (Aer)	-	10 [144]	-	-
PP				
Fixed Angle*	-	10 [144]	-	10 [40]
Fixed Angle [†]	-	10 [144]	-	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp*	-	10 [144]	10 [100]	-
Param. Transfer	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	20 [105–144]	10 [40]	10 [40]
TQA*	-	20 [105–144]	10 [40]	10 [40]
SV				
Fixed Angle*	39 [10–20]	-	49 [10–19]	115 [10–20]
Fixed Angle [†]	39 [10–20]	-	50 [10–19]	121 [10–20]
Fourier*	58 [10–19]	10 [12]	699 [10–20]	266 [10–20]
Interp.	82 [10–20]	10 [12]	693 [10–20]	372 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	81 [10–20]	10 [12]	276 [10–20]	348 [10–20]
TQA	83 [10–20]	10 [12]	659 [10–20]	359 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	95.0 ± 0.5
Fixed Angle* (Aer)	-	84.3 ± 3.3	-	-
Fixed Angle [†]	-	-	-	93.8 ± 0.6
Fixed Angle [†] (Aer)	-	83.3 ± 3.4	-	-
Fourier (Aer)	82.1 ± 6.3	82.4 ± 3.3	80.0 ± 9.5	80.5 ± 4.8
Fourier*	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	90.0 ± 8.3
Interp.	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	93.4 ± 3.6
Interp. (Aer)	91.6 ± 1.6	87.2 ± 3.8	91.0 ± 3.6	86.3 ± 4.7
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	85.7 ± 1.0	64.5 ± 2.1	-
Recursive TS	-	71.1	-	83.7 ± 7.3
TQA	-	73.4 ± 4.6	88.2 ± 1.2	82.6 ± 1.5
TQA (Aer)	-	79.2 ± 1.2	-	-
TQA*	-	75.2 ± 4.8	91.4 ± 1.0	90.2 ± 1.4
TQA* (Aer)	-	81.5 ± 1.3	-	-
PP				
Fixed Angle*	-	90.5 ± 0.8	-	91.7 ± 1.5
Fixed Angle [†]	-	88.7 ± 1.8	-	91.5 ± 1.5
Fourier*	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	89.0 ± 1.5
Interp.	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.5 ± 1.8
Linear Ramp*	-	88.8 ± 0.8	86.9 ± 1.3	-
Param. Transfer	-	-	-	-
Recursive TS	-	88.2	-	89.0 ± 2.3
TQA	-	83.8 ± 1.1	68.6 ± 1.7	85.9 ± 1.4
TQA*	-	88.2 ± 0.8	83.5 ± 2.0	89.7 ± 1.5
SV				
Fixed Angle*	93.2 ± 1.5	-	95.2 ± 1.7	95.3 ± 2.0
Fixed Angle [†]	90.0 ± 2.3	-	94.0 ± 1.8	94.9 ± 2.0
Fourier*	80.0 ± 10.3	79.2 ± 6.8	81.4 ± 8.3	85.2 ± 10.7
Interp.	93.0 ± 1.5	92.5 ± 0.9	92.6 ± 4.1	92.5 ± 4.4
Linear Ramp*	-	-	-	-
Recursive TS	91.9 ± 2.1	92.6 ± 1.3	89.8 ± 3.5	91.1 ± 3.5
TQA	86.7 ± 3.7	83.7 ± 4.9	81.9 ± 6.4	86.6 ± 5.3
TQA*	91.6 ± 1.5	90.7 ± 2.0	90.6 ± 3.4	93.1 ± 2.2

Table 14: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
MPS								
Fixed Angle*	-	-	-	95.0 ± 0.5	-	-	-	
Fixed Angle* (Aer)	-	84.3 ± 3.3	-	-	-	10 [144]	-	
Fixed Angle†	-	-	-	93.8 ± 0.6	-	-	-	
Fixed Angle† (Aer)	-	83.3 ± 3.4	-	-	-	10 [144]	-	
Fourier (Aer)	82.1 ± 6.3	82.4 ± 3.3	80.0 ± 9.5	80.5 ± 4.8	10 [40]	20 [39–144]	30 [40–100]	
Fourier*	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	90.0 ± 8.3	20 [40–50]	31 [39–144]	130 [40–100]	2
Interp.	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	93.4 ± 3.6	20 [40–50]	31 [39–144]	130 [40–100]	2
Interp. (Aer)	91.6 ± 1.6	87.2 ± 3.8	91.0 ± 3.6	86.3 ± 4.7	10 [40]	20 [39–144]	30 [40–100]	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp* (Aer)	-	85.7 ± 1.0	64.5 ± 2.1	-	-	10 [144]	10 [100]	
Recursive TS	-	71.1	-	83.7 ± 7.3	-	1 [106]	-	2
TQA	-	73.4 ± 4.6	88.2 ± 1.2	82.6 ± 1.5	-	20 [105–144]	10 [40]	
TQA (Aer)	-	79.2 ± 1.2	-	-	-	10 [144]	-	
TQA*	-	75.2 ± 4.8	91.4 ± 1.0	90.2 ± 1.4	-	20 [105–144]	10 [40]	
TQA* (Aer)	-	81.5 ± 1.3	-	-	-	10 [144]	-	
PP								
Fixed Angle*	-	90.5 ± 0.8	-	91.7 ± 1.5	-	10 [144]	-	
Fixed Angle†	-	88.7 ± 1.8	-	91.5 ± 1.5	-	10 [144]	-	
Fourier*	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	89.0 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	2
Interp.	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.5 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	2
Linear Ramp*	-	88.8 ± 0.8	86.9 ± 1.3	-	-	10 [144]	10 [100]	
Param. Transfer	-	-	-	-	-	-	-	
Recursive TS	-	88.2	-	89.0 ± 2.3	-	1 [106]	-	2
TQA	-	83.8 ± 1.1	68.6 ± 1.7	85.9 ± 1.4	-	20 [105–144]	10 [40]	
TQA*	-	88.2 ± 0.8	83.5 ± 2.0	89.7 ± 1.5	-	20 [105–144]	10 [40]	
SV								
Fixed Angle*	93.2 ± 1.5	-	95.2 ± 1.7	95.3 ± 2.0	39 [10–20]	-	49 [10–19]	1
Fixed Angle†	90.0 ± 2.3	-	94.0 ± 1.8	94.9 ± 2.0	39 [10–20]	-	50 [10–19]	1
Fourier*	80.0 ± 10.3	79.2 ± 6.8	81.4 ± 8.3	85.2 ± 10.7	58 [10–19]	10 [12]	699 [10–20]	2
Interp.	93.0 ± 1.5	92.5 ± 0.9	92.6 ± 4.1	92.5 ± 4.4	82 [10–20]	10 [12]	693 [10–20]	3
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	91.9 ± 2.1	92.6 ± 1.3	89.8 ± 3.5	91.1 ± 3.5	81 [10–20]	10 [12]	276 [10–20]	3
TQA	86.7 ± 3.7	83.7 ± 4.9	81.9 ± 6.4	86.6 ± 5.3	83 [10–20]	10 [12]	659 [10–20]	3
TQA*	91.6 ± 1.5	90.7 ± 2.0	90.6 ± 3.4	93.1 ± 2.2	81 [10–20]	10 [12]	653 [10–20]	1

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

6 Tables for depth $P = 6$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	10 [100]	10 [40]
Fixed Angle* (Aer)	-	-	20 [100]	10 [40]
Fixed Angle [†]	-	-	10 [100]	10 [40]
Fixed Angle [†] (Aer)	-	-	20 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	30 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
TQA*	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA* (Aer)	10 [40]	10 [39]	30 [40–100]	10 [40]
PP				
Fixed Angle*	-	-	10 [100]	10 [40]
Fixed Angle [†]	-	-	10 [100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp*	-	-	10 [100]	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	130 [40–100]	20 [40]
SV				
Fixed Angle*	16 [11–17]	-	19 [10–18]	59 [10–20]
Fixed Angle [†]	16 [11–17]	-	19 [10–18]	59 [10–20]
Fourier*	56 [10–19]	10 [12]	693 [10–20]	266 [10–20]
Interp.	82 [10–20]	10 [12]	686 [10–20]	372 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	80 [10–20]	10 [12]	269 [10–20]	348 [10–20]
TQA	83 [10–20]	10 [12]	659 [10–20]	360 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]

Table 16: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	95.1 ± 1.1	96.9 ± 0.6
Fixed Angle* (Aer)	-	-	89.9 ± 2.5	97.2 ± 0.2
Fixed Angle [†]	-	-	95.1 ± 1.0	96.1 ± 0.5
Fixed Angle [†] (Aer)	-	-	85.7 ± 5.1	96.6 ± 0.5
Fourier (Aer)	79.3 ± 7.1	83.1 ± 3.0	80.7 ± 8.8	86.1 ± 3.2
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	92.2 ± 6.7
Interp.	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	94.3 ± 3.6
Interp. (Aer)	92.8 ± 1.3	88.3 ± 4.4	93.4 ± 2.5	89.3 ± 3.4
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	65.4 ± 3.0	-
Recursive TS	-	-	-	-
TQA	71.7 ± 6.0	85.6 ± 1.4	88.0 ± 2.1	89.9 ± 2.7
TQA (Aer)	74.2 ± 2.8	84.8 ± 0.9	78.6 ± 8.4	86.3 ± 2.3
TQA*	85.5 ± 6.7	88.8 ± 0.9	91.1 ± 1.4	93.9 ± 1.3
TQA* (Aer)	83.3 ± 3.8	86.8 ± 1.3	84.6 ± 8.7	93.2 ± 1.1
PP				
Fixed Angle*	-	-	89.1 ± 1.8	92.5 ± 1.5
Fixed Angle [†]	-	-	85.5 ± 1.6	92.2 ± 1.6
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.7 ± 1.8
Interp.	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.6 ± 1.9
Linear Ramp*	-	-	87.9 ± 1.3	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	84.6 ± 1.9	85.6 ± 1.4	76.5 ± 5.9	88.0 ± 1.8
TQA*	91.3 ± 2.4	89.8 ± 1.0	85.2 ± 2.2	90.7 ± 1.6
SV				
Fixed Angle*	94.5 ± 1.5	-	96.6 ± 1.4	96.2 ± 1.4
Fixed Angle [†]	90.5 ± 2.5	-	95.7 ± 1.3	95.9 ± 1.3
Fourier*	79.8 ± 11.1	81.3 ± 11.1	81.1 ± 9.4	84.7 ± 11.8
Interp.	94.2 ± 1.3	93.6 ± 1.4	93.8 ± 4.0	93.1 ± 4.9
Linear Ramp*	-	-	-	-
Recursive TS	93.1 ± 1.9	93.1 ± 1.3	91.0 ± 2.7	92.2 ± 3.4
TQA	88.1 ± 3.5	85.6 ± 4.4	82.4 ± 7.3	88.4 ± 5.0
TQA*	93.0 ± 1.4	91.2 ± 2.2	91.9 ± 3.1	94.2 ± 2.2

Table 17: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fixed Angle*	-	-	95.1 ± 1.1	96.9 ± 0.6	-	-	10 [100]
Fixed Angle* (Aer)	-	-	89.9 ± 2.5	97.2 ± 0.2	-	-	20 [100]
Fixed Angle [†]	-	-	95.1 ± 1.0	96.1 ± 0.5	-	-	10 [100]
Fixed Angle [†] (Aer)	-	-	85.7 ± 5.1	96.6 ± 0.5	-	-	20 [100]
Fourier (Aer)	79.3 ± 7.1	83.1 ± 3.0	80.7 ± 8.8	86.1 ± 3.2	10 [40]	20 [39–144]	30 [40–100]
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	92.2 ± 6.7	20 [40–50]	31 [39–144]	130 [40–100]
Interp.	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	94.3 ± 3.6	20 [40–50]	31 [39–144]	130 [40–100]
Interp. (Aer)	92.8 ± 1.3	88.3 ± 4.4	93.4 ± 2.5	89.3 ± 3.4	10 [40]	20 [39–144]	30 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	65.4 ± 3.0	-	-	-	10 [100]
Recursive TS	-	-	-	-	-	-	-
TQA	71.7 ± 6.0	85.6 ± 1.4	88.0 ± 2.1	89.9 ± 2.7	20 [40–50]	10 [39]	130 [40–100]
TQA (Aer)	74.2 ± 2.8	84.8 ± 0.9	78.6 ± 8.4	86.3 ± 2.3	10 [40]	10 [39]	30 [40–100]
TQA*	85.5 ± 6.7	88.8 ± 0.9	91.1 ± 1.4	93.9 ± 1.3	20 [40–50]	10 [39]	130 [40–100]
TQA* (Aer)	83.3 ± 3.8	86.8 ± 1.3	84.6 ± 8.7	93.2 ± 1.1	10 [40]	10 [39]	30 [40–100]
PP							
Fixed Angle*	-	-	89.1 ± 1.8	92.5 ± 1.5	-	-	10 [100]
Fixed Angle [†]	-	-	85.5 ± 1.6	92.2 ± 1.6	-	-	10 [100]
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.7 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]
Interp.	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.6 ± 1.9	20 [40–50]	31 [39–144]	130 [40–100]
Linear Ramp*	-	-	87.9 ± 1.3	-	-	-	10 [100]
Param. Transfer	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	84.6 ± 1.9	85.6 ± 1.4	76.5 ± 5.9	88.0 ± 1.8	20 [40–50]	10 [39]	130 [40–100]
TQA*	91.3 ± 2.4	89.8 ± 1.0	85.2 ± 2.2	90.7 ± 1.6	20 [40–50]	10 [39]	130 [40–100]
SV							
Fixed Angle*	94.5 ± 1.5	-	96.6 ± 1.4	96.2 ± 1.4	16 [11–17]	-	19 [10–18]
Fixed Angle [†]	90.5 ± 2.5	-	95.7 ± 1.3	95.9 ± 1.3	16 [11–17]	-	19 [10–18]
Fourier*	79.8 ± 11.1	81.3 ± 11.1	81.1 ± 9.4	84.7 ± 11.8	56 [10–19]	10 [12]	693 [10–20]
Interp.	94.2 ± 1.3	93.6 ± 1.4	93.8 ± 4.0	93.1 ± 4.9	82 [10–20]	10 [12]	686 [10–20]
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	93.1 ± 1.9	93.1 ± 1.3	91.0 ± 2.7	92.2 ± 3.4	80 [10–20]	10 [12]	269 [10–20]
TQA	88.1 ± 3.5	85.6 ± 4.4	82.4 ± 7.3	88.4 ± 5.0	83 [10–20]	10 [12]	659 [10–20]
TQA*	93.0 ± 1.4	91.2 ± 2.2	91.9 ± 3.1	94.2 ± 2.2	81 [10–20]	10 [12]	653 [10–20]

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

7 Tables for depth $P = 7$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA (Aer)	-	-	-	-
TQA*	-	-	-	-
TQA* (Aer)	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	16 [11–17]	-	18 [10–18]	59 [10–20]
Fixed Angle [†]	16 [11–17]	-	18 [10–18]	60 [10–20]
Fourier*	54 [10–18]	10 [12]	688 [10–20]	266 [10–20]
Interp.	82 [10–20]	10 [12]	681 [10–20]	372 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	80 [10–20]	10 [12]	270 [10–20]	349 [10–20]
TQA	83 [10–20]	10 [12]	659 [10–20]	360 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	135 [10–20]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle [*]	-	-	-	-
Fixed Angle [*] (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	81.3 ± 6.9	84.4 ± 3.2	81.9 ± 10.0	88.2 ± 6.1
Fourier [*]	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	91.0 ± 7.6
Interp.	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	95.1 ± 3.9
Interp. (Aer)	94.0 ± 1.5	89.2 ± 4.7	95.0 ± 2.3	90.4 ± 2.4
Linear Ramp [*]	-	-	-	-
Linear Ramp [*] (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA (Aer)	-	-	-	-
TQA [*]	-	-	-	-
TQA [*] (Aer)	-	-	-	-
PP				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	90.1 ± 1.8
Interp.	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.5 ± 2.0
Linear Ramp [*]	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
SV				
Fixed Angle [*]	95.5 ± 1.6	-	97.5 ± 1.1	97.2 ± 1.2
Fixed Angle [†]	91.0 ± 3.0	-	96.4 ± 1.0	96.8 ± 1.2
Fourier [*]	80.0 ± 10.9	81.7 ± 10.2	81.0 ± 9.8	84.7 ± 12.2
Interp.	95.1 ± 1.3	94.3 ± 1.2	94.7 ± 4.2	93.3 ± 5.4
Linear Ramp [*]	-	-	-	-
Recursive TS	93.9 ± 1.9	93.8 ± 1.0	92.4 ± 4.1	92.9 ± 3.4
TQA	89.2 ± 3.4	87.0 ± 4.1	83.0 ± 7.8	89.3 ± 4.8
TQA [*]	93.9 ± 1.4	91.8 ± 2.2	92.5 ± 3.2	95.1 ± 1.8

Table 20: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-	-	-	-
Fourier (Aer)	81.3 ± 6.9	84.4 ± 3.2	81.9 ± 10.0	88.2 ± 6.1	10 [40]	20 [39–144]	20 [40–100]
Fourier*	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	91.0 ± 7.6	20 [40–50]	31 [39–144]	120 [40–100]
Interp.	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	95.1 ± 3.9	20 [40–50]	31 [39–144]	120 [40–100]
Interp. (Aer)	94.0 ± 1.5	89.2 ± 4.7	95.0 ± 2.3	90.4 ± 2.4	10 [40]	20 [39–144]	20 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
TQA* (Aer)	-	-	-	-	-	-	-
PP							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fourier*	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	90.1 ± 1.8	20 [40–50]	31 [39–144]	120 [40–100]
Interp.	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.5 ± 2.0	20 [40–50]	31 [39–144]	120 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Param. Transfer	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
SV							
Fixed Angle*	95.5 ± 1.6	-	97.5 ± 1.1	97.2 ± 1.2	16 [11–17]	-	18 [10–18]
Fixed Angle [†]	91.0 ± 3.0	-	96.4 ± 1.0	96.8 ± 1.2	16 [11–17]	-	18 [10–18]
Fourier*	80.0 ± 10.9	81.7 ± 10.2	81.0 ± 9.8	84.7 ± 12.2	54 [10–18]	10 [12]	688 [10–20]
Interp.	95.1 ± 1.3	94.3 ± 1.2	94.7 ± 4.2	93.3 ± 5.4	82 [10–20]	10 [12]	681 [10–20]
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	93.9 ± 1.9	93.8 ± 1.0	92.4 ± 4.1	92.9 ± 3.4	80 [10–20]	10 [12]	270 [10–20]
TQA	89.2 ± 3.4	87.0 ± 4.1	83.0 ± 7.8	89.3 ± 4.8	83 [10–20]	10 [12]	659 [10–20]
TQA*	93.9 ± 1.4	91.8 ± 2.2	92.5 ± 3.2	95.1 ± 1.8	81 [10–20]	10 [12]	653 [10–20]

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

8 Tables for depth $P = 8$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	10 [100]	10 [40]
Fixed Angle* (Aer)	-	-	10 [100]	10 [40]
Fixed Angle [†]	-	-	10 [100]	10 [40]
Fixed Angle [†] (Aer)	-	-	10 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	20 [40–100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA (Aer)	10 [40]	10 [39]	20 [40–100]	10 [40]
TQA*	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA* (Aer)	10 [40]	10 [39]	20 [40–100]	10 [40]
PP				
Fixed Angle*	-	-	10 [100]	10 [40]
Fixed Angle [†]	-	-	10 [100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp.	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	120 [40–100]	20 [40]
SV				
Fixed Angle*	16 [11–17]	-	18 [10–18]	50 [10–20]
Fixed Angle [†]	16 [11–17]	-	18 [10–18]	50 [10–20]
Fourier*	53 [10–18]	10 [12]	681 [10–20]	266 [10–20]
Interp.	82 [10–20]	10 [12]	676 [10–20]	372 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	80 [10–20]	10 [12]	265 [10–20]	348 [10–20]
TQA	83 [10–20]	10 [12]	659 [10–20]	355 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]

Table 22: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	96.7 ± 0.8	98.3 ± 0.3
Fixed Angle* (Aer)	-	-	93.6 ± 0.8	98.4 ± 0.5
Fixed Angle [†]	-	-	96.6 ± 0.8	97.9 ± 0.4
Fixed Angle [†] (Aer)	-	-	89.5 ± 1.9	98.1 ± 0.3
Fourier (Aer)	79.5 ± 7.1	85.2 ± 3.2	82.9 ± 9.6	86.5 ± 8.7
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	93.8 ± 6.3
Interp.	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	96.1 ± 3.2
Interp. (Aer)	94.3 ± 1.5	90.0 ± 4.7	96.2 ± 1.6	90.8 ± 2.7
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	82.2 ± 6.9	88.3 ± 1.2	89.3 ± 2.0	93.5 ± 2.0
TQA (Aer)	79.5 ± 3.5	87.5 ± 0.8	80.9 ± 7.2	91.1 ± 1.4
TQA*	91.1 ± 3.7	90.3 ± 1.0	92.2 ± 1.2	96.4 ± 1.0
TQA* (Aer)	86.3 ± 1.7	89.3 ± 1.0	88.2 ± 5.9	96.0 ± 0.5
PP				
Fixed Angle*	-	-	90.3 ± 1.9	93.4 ± 1.6
Fixed Angle [†]	-	-	86.2 ± 1.7	92.8 ± 1.6
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.4 ± 2.2
Interp.	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.9 ± 2.1
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.2	77.4 ± 6.7	89.9 ± 1.6
TQA*	91.7 ± 2.3	91.1 ± 0.7	85.6 ± 2.5	92.0 ± 1.5
SV				
Fixed Angle*	96.4 ± 1.4	-	98.1 ± 0.9	97.8 ± 1.1
Fixed Angle [†]	91.8 ± 3.2	-	96.8 ± 1.1	97.2 ± 1.1
Fourier*	79.0 ± 12.3	79.4 ± 11.1	79.7 ± 11.2	84.1 ± 12.8
Interp.	95.8 ± 1.2	94.7 ± 1.1	95.3 ± 4.0	93.7 ± 5.4
Linear Ramp*	-	-	-	-
Recursive TS	94.6 ± 1.8	94.3 ± 1.0	93.3 ± 4.6	93.4 ± 3.3
TQA	90.2 ± 3.5	88.1 ± 4.0	83.5 ± 8.5	90.0 ± 4.9
TQA*	94.4 ± 1.3	92.3 ± 1.8	93.2 ± 3.3	95.7 ± 1.6

Table 23: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fixed Angle*	-	-	96.7 ± 0.8	98.3 ± 0.3	-	-	10 [100]
Fixed Angle* (Aer)	-	-	93.6 ± 0.8	98.4 ± 0.5	-	-	10 [100]
Fixed Angle [†]	-	-	96.6 ± 0.8	97.9 ± 0.4	-	-	10 [100]
Fixed Angle [†] (Aer)	-	-	89.5 ± 1.9	98.1 ± 0.3	-	-	10 [100]
Fourier (Aer)	79.5 ± 7.1	85.2 ± 3.2	82.9 ± 9.6	86.5 ± 8.7	10 [40]	20 [39–144]	20 [40–100]
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	93.8 ± 6.3	20 [40–50]	31 [39–144]	120 [40–100]
Interp.	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	96.1 ± 3.2	20 [40–50]	31 [39–144]	120 [40–100]
Interp. (Aer)	94.3 ± 1.5	90.0 ± 4.7	96.2 ± 1.6	90.8 ± 2.7	10 [40]	20 [39–144]	20 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	82.2 ± 6.9	88.3 ± 1.2	89.3 ± 2.0	93.5 ± 2.0	20 [40–50]	10 [39]	120 [40–100]
TQA (Aer)	79.5 ± 3.5	87.5 ± 0.8	80.9 ± 7.2	91.1 ± 1.4	10 [40]	10 [39]	20 [40–100]
TQA*	91.1 ± 3.7	90.3 ± 1.0	92.2 ± 1.2	96.4 ± 1.0	20 [40–50]	10 [39]	120 [40–100]
TQA* (Aer)	86.3 ± 1.7	89.3 ± 1.0	88.2 ± 5.9	96.0 ± 0.5	10 [40]	10 [39]	20 [40–100]
PP							
Fixed Angle*	-	-	90.3 ± 1.9	93.4 ± 1.6	-	-	10 [100]
Fixed Angle [†]	-	-	86.2 ± 1.7	92.8 ± 1.6	-	-	10 [100]
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.4 ± 2.2	20 [40–50]	31 [39–144]	120 [40–100]
Interp.	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.9 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Param. Transfer	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.2	77.4 ± 6.7	89.9 ± 1.6	20 [40–50]	10 [39]	120 [40–100]
TQA*	91.7 ± 2.3	91.1 ± 0.7	85.6 ± 2.5	92.0 ± 1.5	20 [40–50]	10 [39]	120 [40–100]
SV							
Fixed Angle*	96.4 ± 1.4	-	98.1 ± 0.9	97.8 ± 1.1	16 [11–17]	-	18 [10–18]
Fixed Angle [†]	91.8 ± 3.2	-	96.8 ± 1.1	97.2 ± 1.1	16 [11–17]	-	18 [10–18]
Fourier*	79.0 ± 12.3	79.4 ± 11.1	79.7 ± 11.2	84.1 ± 12.8	53 [10–18]	10 [12]	681 [10–20]
Interp.	95.8 ± 1.2	94.7 ± 1.1	95.3 ± 4.0	93.7 ± 5.4	82 [10–20]	10 [12]	676 [10–20]
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	94.6 ± 1.8	94.3 ± 1.0	93.3 ± 4.6	93.4 ± 3.3	80 [10–20]	10 [12]	265 [10–20]
TQA	90.2 ± 3.5	88.1 ± 4.0	83.5 ± 8.5	90.0 ± 4.9	83 [10–20]	10 [12]	659 [10–20]
TQA*	94.4 ± 1.3	92.3 ± 1.8	93.2 ± 3.3	95.7 ± 1.6	81 [10–20]	10 [12]	653 [10–20]

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

9 Tables for depth $P = 9$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp.	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Linear Ramp*	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA (Aer)	-	-	-	-
TQA*	-	-	-	-
TQA* (Aer)	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp.	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Linear Ramp*	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	16 [11–17]	-	18 [10–18]	50 [10–20]
Fixed Angle [†]	16 [11–17]	-	18 [10–18]	50 [10–20]
Fourier*	52 [10–18]	10 [12]	672 [10–20]	264 [10–20]
Interp.	82 [10–20]	10 [12]	673 [10–20]	371 [10–20]
Linear Ramp*	-	-	-	-
Recursive TS	80 [10–20]	10 [12]	264 [10–20]	352 [10–20]
TQA	83 [10–20]	10 [12]	658 [10–20]	356 [10–20]
TQA*	81 [10–20]	10 [12]	652 [10–20]	135 [10–20]

Table 25: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle [*]	-	-	-	-
Fixed Angle [*] (Aer)	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-
Fourier (Aer)	79.0 ± 6.7	85.8 ± 4.0	73.6 ± 7.4	89.7 ± 3.2
Fourier [*]	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	92.7 ± 7.4
Interp.	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	93.1 ± 4.5
Interp. (Aer)	94.1 ± 4.2	90.6 ± 4.8	95.9 ± 1.7	93.5 ± 1.9
Linear Ramp [*]	-	-	-	-
Linear Ramp [*] (Aer)	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA (Aer)	-	-	-	-
TQA [*]	-	-	-	-
TQA [*] (Aer)	-	-	-	-
PP				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	91.0 ± 2.6
Interp.	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3
Linear Ramp [*]	-	-	-	-
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
SV				
Fixed Angle [*]	97.0 ± 1.4	-	98.5 ± 0.8	98.3 ± 1.1
Fixed Angle [†]	92.1 ± 3.4	-	97.0 ± 1.3	97.7 ± 1.0
Fourier [*]	78.2 ± 12.7	78.0 ± 10.8	79.1 ± 11.6	83.5 ± 13.7
Interp.	96.3 ± 1.1	94.8 ± 1.2	95.7 ± 4.3	93.6 ± 6.1
Linear Ramp [*]	-	-	-	-
Recursive TS	95.0 ± 1.9	94.7 ± 0.9	93.6 ± 5.2	93.6 ± 3.2
TQA	90.9 ± 3.7	89.1 ± 3.8	83.7 ± 9.2	90.8 ± 4.9
TQA [*]	95.0 ± 1.3	93.0 ± 1.6	93.6 ± 3.3	96.2 ± 1.7

Table 26: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-	-	-	-
Fourier (Aer)	79.0 ± 6.7	85.8 ± 4.0	73.6 ± 7.4	89.7 ± 3.2	10 [40]	20 [39–144]	10 [100]
Fourier*	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	92.7 ± 7.4	20 [40–50]	31 [39–144]	111 [40–100]
Interp.	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	93.1 ± 4.5	20 [40–50]	31 [39–144]	111 [40–100]
Interp. (Aer)	94.1 ± 4.2	90.6 ± 4.8	95.9 ± 1.7	93.5 ± 1.9	10 [40]	20 [39–144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
TQA* (Aer)	-	-	-	-	-	-	-
PP							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fourier*	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	91.0 ± 2.6	20 [40–50]	31 [39–144]	111 [40–100]
Interp.	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Param. Transfer	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
SV							
Fixed Angle*	97.0 ± 1.4	-	98.5 ± 0.8	98.3 ± 1.1	16 [11–17]	-	18 [10–18]
Fixed Angle [†]	92.1 ± 3.4	-	97.0 ± 1.3	97.7 ± 1.0	16 [11–17]	-	18 [10–18]
Fourier*	78.2 ± 12.7	78.0 ± 10.8	79.1 ± 11.6	83.5 ± 13.7	52 [10–18]	10 [12]	672 [10–20]
Interp.	96.3 ± 1.1	94.8 ± 1.2	95.7 ± 4.3	93.6 ± 6.1	82 [10–20]	10 [12]	673 [10–20]
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	95.0 ± 1.9	94.7 ± 0.9	93.6 ± 5.2	93.6 ± 3.2	80 [10–20]	10 [12]	264 [10–20]
TQA	90.9 ± 3.7	89.1 ± 3.8	83.7 ± 9.2	90.8 ± 4.9	83 [10–20]	10 [12]	658 [10–20]
TQA*	95.0 ± 1.3	93.0 ± 1.6	93.6 ± 3.3	96.2 ± 1.7	81 [10–20]	10 [12]	652 [10–20]

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

10 Tables for depth $P = 10$

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	10 [100]	11 [40–100]
Fixed Angle* (Aer)	-	10 [144]	10 [100]	10 [40]
Fixed Angle [†]	-	-	10 [100]	11 [40–100]
Fixed Angle [†] (Aer)	-	10 [144]	10 [100]	10 [40]
Fourier (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp.	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp. (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
Linear Ramp*	10 [40]	10 [39]	-	-
Linear Ramp* (Aer)	10 [40]	30 [39–144]	44 [40–100]	10 [40]
Recursive TS	-	-	-	-
TQA	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
TQA (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
TQA*	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
TQA* (Aer)	10 [40]	20 [39–144]	10 [100]	10 [40]
PP				
Fixed Angle*	-	10 [144]	10 [100]	11 [40–100]
Fixed Angle [†]	-	10 [144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp.	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Linear Ramp*	10 [40]	20 [39–144]	118 [40–100]	10 [40]
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	30 [39–144]	111 [40–100]	11 [40–100]
TQA*	20 [40–50]	30 [39–144]	111 [40–100]	11 [40–100]
SV				
Fixed Angle*	16 [11–17]	-	18 [10–18]	43 [10–22]
Fixed Angle [†]	16 [11–17]	-	18 [10–18]	46 [10–24]
Fourier*	50 [10–17]	10 [12]	665 [10–20]	252 [10–20]
Interp.	82 [10–20]	10 [12]	664 [10–20]	353 [10–20]
Linear Ramp*	88 [10–20]	10 [12]	175 [10–20]	568 [10–20]
Recursive TS	79 [10–20]	9 [12]	264 [10–20]	333 [10–20]
TQA	83 [10–20]	10 [12]	657 [10–20]	349 [10–22]
TQA*	81 [10–20]	9 [12]	651 [10–20]	147 [10–20]

Table 28: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fixed Angle*	-	-	97.6 ± 0.7	98.6 ± 0.9
Fixed Angle* (Aer)	-	86.3 ± 2.4	96.0 ± 0.8	99.0 ± 0.4
Fixed Angle [†]	-	-	97.4 ± 0.7	98.2 ± 1.4
Fixed Angle [†] (Aer)	-	85.3 ± 2.2	93.4 ± 1.7	98.8 ± 0.3
Fourier (Aer)	82.5 ± 7.8	85.1 ± 4.8	77.1 ± 7.3	91.6 ± 3.7
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.7 ± 5.6	92.8 ± 9.0
Interp.	97.4 ± 1.5	86.6 ± 10.1	96.6 ± 0.9	93.8 ± 4.6
Interp. (Aer)	94.3 ± 4.5	91.3 ± 4.6	96.9 ± 1.0	92.0 ± 1.6
Linear Ramp*	96.8 ± 0.5	93.6 ± 0.7	-	-
Linear Ramp* (Aer)	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8
Recursive TS	-	-	-	-
TQA	87.7 ± 1.8	79.6 ± 9.2	90.0 ± 2.5	94.0 ± 3.3
TQA (Aer)	85.9 ± 2.1	87.1 ± 2.5	83.7 ± 2.6	94.1 ± 0.9
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.6 ± 1.7	96.5 ± 1.7
TQA* (Aer)	90.1 ± 1.9	87.9 ± 2.9	88.3 ± 2.3	96.5 ± 0.6
PP				
Fixed Angle*	-	94.0 ± 0.5	90.7 ± 1.9	93.6 ± 1.6
Fixed Angle [†]	-	91.1 ± 1.4	86.7 ± 1.7	93.0 ± 1.6
Fourier*	91.5 ± 1.9	91.1 ± 3.2	86.0 ± 2.2	91.3 ± 2.6
Interp.	94.4 ± 1.9	94.9 ± 0.8	89.6 ± 1.7	90.1 ± 2.4
Linear Ramp*	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6
Param. Transfer	-	-	-	-
Recursive TS	-	-	-	-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	90.8 ± 1.6
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	92.3 ± 1.6
SV				
Fixed Angle*	97.4 ± 1.4	-	98.8 ± 0.6	98.7 ± 0.9
Fixed Angle [†]	92.4 ± 3.5	-	97.2 ± 1.3	98.1 ± 0.9
Fourier*	77.5 ± 13.2	78.5 ± 11.7	77.9 ± 12.3	83.4 ± 13.9
Interp.	96.6 ± 1.1	94.8 ± 1.1	96.0 ± 3.9	94.0 ± 5.9
Linear Ramp*	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8
Recursive TS	95.3 ± 2.0	95.2 ± 0.8	93.4 ± 4.4	94.0 ± 3.2
TQA	91.3 ± 3.9	89.9 ± 3.6	83.8 ± 9.5	91.0 ± 5.1
TQA*	95.4 ± 1.4	93.2 ± 2.1	93.9 ± 3.2	96.7 ± 1.5

Table 29: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fixed Angle*	-	-	97.6 ± 0.7	98.6 ± 0.9	-	-	10 [100]
Fixed Angle* (Aer)	-	86.3 ± 2.4	96.0 ± 0.8	99.0 ± 0.4	-	10 [144]	10 [100]
Fixed Angle [†]	-	-	97.4 ± 0.7	98.2 ± 1.4	-	-	10 [100]
Fixed Angle [†] (Aer)	-	85.3 ± 2.2	93.4 ± 1.7	98.8 ± 0.3	-	10 [144]	10 [100]
Fourier (Aer)	82.5 ± 7.8	85.1 ± 4.8	77.1 ± 7.3	91.6 ± 3.7	10 [40]	20 [39–144]	10 [100]
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.7 ± 5.6	92.8 ± 9.0	20 [40–50]	31 [39–144]	111 [40–100]
Interp.	97.4 ± 1.5	86.6 ± 10.1	96.6 ± 0.9	93.8 ± 4.6	20 [40–50]	31 [39–144]	111 [40–100]
Interp. (Aer)	94.3 ± 4.5	91.3 ± 4.6	96.9 ± 1.0	92.0 ± 1.6	10 [40]	20 [39–144]	10 [100]
Linear Ramp*	96.8 ± 0.5	93.6 ± 0.7	-	-	10 [40]	10 [39]	-
Linear Ramp* (Aer)	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8	10 [40]	30 [39–144]	44 [40–100]
Recursive TS	-	-	-	-	-	-	-
TQA	87.7 ± 1.8	79.6 ± 9.2	90.0 ± 2.5	94.0 ± 3.3	20 [40–50]	31 [39–144]	111 [40–100]
TQA (Aer)	85.9 ± 2.1	87.1 ± 2.5	83.7 ± 2.6	94.1 ± 0.9	10 [40]	20 [39–144]	10 [100]
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.6 ± 1.7	96.5 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]
TQA* (Aer)	90.1 ± 1.9	87.9 ± 2.9	88.3 ± 2.3	96.5 ± 0.6	10 [40]	20 [39–144]	10 [100]
PP							
Fixed Angle*	-	94.0 ± 0.5	90.7 ± 1.9	93.6 ± 1.6	-	10 [144]	10 [100]
Fixed Angle [†]	-	91.1 ± 1.4	86.7 ± 1.7	93.0 ± 1.6	-	10 [144]	10 [100]
Fourier*	91.5 ± 1.9	91.1 ± 3.2	86.0 ± 2.2	91.3 ± 2.6	20 [40–50]	31 [39–144]	111 [40–100]
Interp.	94.4 ± 1.9	94.9 ± 0.8	89.6 ± 1.7	90.1 ± 2.4	20 [40–50]	31 [39–144]	111 [40–100]
Linear Ramp*	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6	10 [40]	20 [39–144]	118 [40–100]
Param. Transfer	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	90.8 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	92.3 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]
SV							
Fixed Angle*	97.4 ± 1.4	-	98.8 ± 0.6	98.7 ± 0.9	16 [11–17]	-	18 [10–18]
Fixed Angle [†]	92.4 ± 3.5	-	97.2 ± 1.3	98.1 ± 0.9	16 [11–17]	-	18 [10–18]
Fourier*	77.5 ± 13.2	78.5 ± 11.7	77.9 ± 12.3	83.4 ± 13.9	50 [10–17]	10 [12]	665 [10–20]
Interp.	96.6 ± 1.1	94.8 ± 1.1	96.0 ± 3.9	94.0 ± 5.9	82 [10–20]	10 [12]	664 [10–20]
Linear Ramp*	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8	88 [10–20]	10 [12]	175 [10–20]
Recursive TS	95.3 ± 2.0	95.2 ± 0.8	93.4 ± 4.4	94.0 ± 3.2	79 [10–20]	9 [12]	264 [10–20]
TQA	91.3 ± 3.9	89.9 ± 3.6	83.8 ± 9.5	91.0 ± 5.1	83 [10–20]	10 [12]	657 [10–20]
TQA*	95.4 ± 1.4	93.2 ± 2.1	93.9 ± 3.2	96.7 ± 1.5	81 [10–20]	9 [12]	651 [10–20]

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

Summary Tables of MIS Data

11 Tables for depth $P = 1$

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	-	-	-
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-	-	-
Fourier (Aer)	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	-	-	-
TQA*	-	-	-
TQA* (Aer)	-	-	-
PP			
Fixed Angle*	-	-	-
Fixed Angle [†]	-	-	-
Fourier*	10 [70]	10 [100]	10 [100]
Interp.	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fixed Angle*	-	-	10 [20]
Fixed Angle [†]	-	-	10 [20]
Fourier*	-	-	10 [20]
Interp.	-	-	10 [20]
Linear Ramp*	-	-	-
Recursive TS	-	-	10 [20]
TQA	-	-	10 [20]
TQA*	-	-	10 [20]

Table 31: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	-	-	-
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-	-	-
Fourier (Aer)	56682.2 ± 2535.8	15317.6 ± 0.0	9229.9 ± 7.4
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	56598.1 ± 2542.4	15317.6 ± 0.0	9229.9 ± 7.4
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	-	-	-
TQA*	-	-	-
TQA* (Aer)	-	-	-
PP			
Fixed Angle*	-	-	-
Fixed Angle [†]	-	-	-
Fourier*	56682.2 ± 2535.8	15317.6 ± 0.0	9229.9 ± 7.4
Interp.	56598.1 ± 2542.4	15317.6 ± 0.0	9229.9 ± 7.4
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fixed Angle*	-	-	1857.0 ± 11.0
Fixed Angle [†]	-	-	1217.6 ± 2.6
Fourier*	-	-	1857.0 ± 11.0
Interp.	-	-	1857.0 ± 11.0
Linear Ramp*	-	-	-
Recursive TS	-	-	1857.0 ± 11.0
TQA	-	-	1789.5 ± 10.7
TQA*	-	-	1857.0 ± 11.0

Table 32: **Avg. Energy ± standard deviation for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	L2F	RR	ER	L2F	RR
MPS						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-	-	-
Fourier (Aer)	56682.2 ± 2535.8	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-	-	-	-
Interp.	-	-	-	-	-	-
Interp. (Aer)	56598.1 ± 2542.4	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
TQA* (Aer)	-	-	-	-	-	-
PP						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-
Fourier*	56682.2 ± 2535.8	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [100]	10 [100]
Interp.	56598.1 ± 2542.4	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
SV						
Fixed Angle*	-	-	1857.0 ± 11.0	-	-	10 [20]
Fixed Angle [†]	-	-	1217.6 ± 2.6	-	-	10 [20]
Fourier*	-	-	1857.0 ± 11.0	-	-	10 [20]
Interp.	-	-	1857.0 ± 11.0	-	-	10 [20]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	1857.0 ± 11.0	-	-	10 [20]
TQA	-	-	1789.5 ± 10.7	-	-	10 [20]
TQA*	-	-	1857.0 ± 11.0	-	-	10 [20]

Table 33: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

12 Tables for depth $P = 2$

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	10 [70]	10 [100]	10 [100]
Fixed Angle†	-	-	-
Fixed Angle† (Aer)	10 [70]	10 [100]	10 [100]
Fourier (Aer)	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	10 [70]	10 [100]	10 [100]
TQA*	-	-	-
TQA* (Aer)	10 [70]	10 [100]	10 [100]
PP			
Fixed Angle*	10 [70]	10 [100]	10 [100]
Fixed Angle†	10 [70]	10 [100]	10 [100]
Fourier*	10 [70]	10 [100]	10 [100]
Interp.	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	10 [70]	10 [100]	10 [100]
TQA*	10 [70]	10 [100]	10 [100]
SV			
Fixed Angle*	-	-	10 [20]
Fixed Angle†	-	-	10 [20]
Fourier*	-	-	10 [20]
Interp.	-	-	10 [20]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	10 [20]
TQA*	-	-	10 [20]

Table 34: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	11603.1 $\pm$ 5689.7	20451.5 $\pm$ 221.4	14594.1 $\pm$ 163.9
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	3358.9 $\pm$ 3731.9	17687.2 $\pm$ 21.1	6117.7 $\pm$ 67.3
Fourier (Aer)	61936.6 $\pm$ 2524.1	23286.0 $\pm$ 577.7	16822.9 $\pm$ 133.1
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	53447.9 $\pm$ 6357.5	19285.1 $\pm$ 523.1	16950.8 $\pm$ 64.1
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	26859.1 $\pm$ 3717.5	21309.1 $\pm$ 17.0	15015.9 $\pm$ 134.0
TQA*	-	-	-
TQA* (Aer)	59220.7 $\pm$ 3629.8	23709.2 $\pm$ 162.9	16962.7 $\pm$ 77.7
PP			
Fixed Angle*	1462.7 $\pm$ 1983.1	19901.6 $\pm$ 0.0	12235.0 $\pm$ 22.2
Fixed Angle [†]	-159.9 $\pm$ 625.3	11104.2 $\pm$ 0.0	6236.1 $\pm$ 12.9
Fourier*	66023.6 $\pm$ 2531.7	23962.1 $\pm$ 0.0	16649.5 $\pm$ 7.4
Interp.	53045.1 $\pm$ 6171.8	20205.4 $\pm$ 0.0	16648.6 $\pm$ 7.9
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	23622.4 $\pm$ 2109.4	21688.0 $\pm$ 0.0	14559.1 $\pm$ 5.9
TQA*	64848.8 $\pm$ 5913.4	23961.1 $\pm$ 0.0	16645.6 $\pm$ 8.6
SV			
Fixed Angle*	-	-	2494.3 $\pm$ 22.7
Fixed Angle [†]	-	-	1289.9 $\pm$ 28.1
Fourier*	-	-	3294.5 $\pm$ 10.2
Interp.	-	-	3294.0 $\pm$ 12.9
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	2912.4 $\pm$ 6.2
TQA*	-	-	3315.8 $\pm$ 8.7

Table 35: **Avg. Energy $\pm$ standard deviation for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	L2F	RR	ER	L2F	RR
MPS						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle* (Aer)	11603.1 $\pm$ 5689.7	20451.5 $\pm$ 221.4	14594.1 $\pm$ 163.9	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-	-	-	-	-	-
Fixed Angle [†] (Aer)	3358.9 $\pm$ 3731.9	17687.2 $\pm$ 21.1	6117.7 $\pm$ 67.3	10 [70]	10 [100]	10 [100]
Fourier (Aer)	61936.6 $\pm$ 2524.1	23286.0 $\pm$ 577.7	16822.9 $\pm$ 133.1	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-	-	-	-
Interp.	-	-	-	-	-	-
Interp. (Aer)	53447.9 $\pm$ 6357.5	19285.1 $\pm$ 523.1	16950.8 $\pm$ 64.1	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA (Aer)	26859.1 $\pm$ 3717.5	21309.1 $\pm$ 17.0	15015.9 $\pm$ 134.0	10 [70]	10 [100]	10 [100]
TQA*	-	-	-	-	-	-
TQA* (Aer)	59220.7 $\pm$ 3629.8	23709.2 $\pm$ 162.9	16962.7 $\pm$ 77.7	10 [70]	10 [100]	10 [100]
PP						
Fixed Angle*	1462.7 $\pm$ 1983.1	19901.6 $\pm$ 0.0	12235.0 $\pm$ 22.2	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-159.9 $\pm$ 625.3	11104.2 $\pm$ 0.0	6236.1 $\pm$ 12.9	10 [70]	10 [100]	10 [100]
Fourier*	66023.6 $\pm$ 2531.7	23962.1 $\pm$ 0.0	16649.5 $\pm$ 7.4	10 [70]	10 [100]	10 [100]
Interp.	53045.1 $\pm$ 6171.8	20205.4 $\pm$ 0.0	16648.6 $\pm$ 7.9	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	23622.4 $\pm$ 2109.4	21688.0 $\pm$ 0.0	14559.1 $\pm$ 5.9	10 [70]	10 [100]	10 [100]
TQA*	64848.8 $\pm$ 5913.4	23961.1 $\pm$ 0.0	16645.6 $\pm$ 8.6	10 [70]	10 [100]	10 [100]
SV						
Fixed Angle*	-	-	2494.3 $\pm$ 22.7	-	-	10 [20]
Fixed Angle [†]	-	-	1289.9 $\pm$ 28.1	-	-	10 [20]
Fourier*	-	-	3294.5 $\pm$ 10.2	-	-	10 [20]
Interp.	-	-	3294.0 $\pm$ 12.9	-	-	10 [20]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	2912.4 $\pm$ 6.2	-	-	10 [20]
TQA*	-	-	3315.8 $\pm$ 8.7	-	-	10 [20]

Table 36: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

13 Tables for depth $P = 3$

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	-	-	-
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-	-	-
Fourier (Aer)	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	-	-	-
TQA*	-	-	-
TQA* (Aer)	-	-	-
PP			
Fixed Angle*	-	-	-
Fixed Angle [†]	-	-	-
Fourier*	10 [70]	10 [100]	10 [100]
Interp.	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fixed Angle*	-	-	10 [20]
Fixed Angle [†]	-	-	10 [20]
Fourier*	-	-	10 [20]
Interp.	-	-	10 [20]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	10 [20]
TQA*	-	-	10 [20]

Table 37: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	L2F	RR
MPS			
Fixed Angle [*]	-	-	-
Fixed Angle [*] (Aer)	-	-	-
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-	-	-
Fourier (Aer)	63673.4 ± 2329.8	23195.5 ± 513.9	17145.6 ± 166.6
Fourier [*]	-	-	-
Interp.	-	-	-
Interp. (Aer)	30291.4 ± 8828.4	20601.2 ± 388.8	17142.0 ± 124.0
Linear Ramp [*]	-	-	-
Linear Ramp [*] (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	-	-	-
TQA [*]	-	-	-
TQA [*] (Aer)	-	-	-
PP			
Fixed Angle [*]	-	-	-
Fixed Angle [†]	-	-	-
Fourier [*]	70899.1 ± 2921.4	25141.6 ± 0.0	17514.9 ± 4.0
Interp.	15399.2 ± 5117.8	23006.9 ± 0.0	17490.7 ± 13.7
Linear Ramp [*]	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA [*]	-	-	-
SV			
Fixed Angle [*]	-	-	3310.9 ± 10.3
Fixed Angle [†]	-	-	929.2 ± 17.4
Fourier [*]	-	-	3335.3 ± 47.3
Interp.	-	-	3232.8 ± 18.9
Linear Ramp [*]	-	-	-
Recursive TS	-	-	-
TQA	-	-	3260.1 ± 9.3
TQA [*]	-	-	3479.4 ± 5.4

Table 38: **Avg. Energy ± standard deviation for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	L2F	RR	ER	L2F	RR
MPS						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-	-	-
Fourier (Aer)	63673.4 $\pm$ 2329.8	23195.5 $\pm$ 513.9	17145.6 $\pm$ 166.6	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-	-	-	-
Interp.	-	-	-	-	-	-
Interp. (Aer)	30291.4 $\pm$ 8828.4	20601.2 $\pm$ 388.8	17142.0 $\pm$ 124.0	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
TQA* (Aer)	-	-	-	-	-	-
PP						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-
Fourier*	70899.1 $\pm$ 2921.4	25141.6 $\pm$ 0.0	17514.9 $\pm$ 4.0	10 [70]	10 [100]	10 [100]
Interp.	15399.2 $\pm$ 5117.8	23006.9 $\pm$ 0.0	17490.7 $\pm$ 13.7	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
SV						
Fixed Angle*	-	-	3310.9 $\pm$ 10.3	-	-	10 [20]
Fixed Angle [†]	-	-	929.2 $\pm$ 17.4	-	-	10 [20]
Fourier*	-	-	3335.3 $\pm$ 47.3	-	-	10 [20]
Interp.	-	-	3232.8 $\pm$ 18.9	-	-	10 [20]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	3260.1 $\pm$ 9.3	-	-	10 [20]
TQA*	-	-	3479.4 $\pm$ 5.4	-	-	10 [20]

Table 39: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

14 Tables for depth $P = 4$

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	10 [70]	10 [100]	10 [100]
Fixed Angle†	-	-	-
Fixed Angle† (Aer)	10 [70]	10 [100]	10 [100]
Fourier (Aer)	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	10 [70]	10 [100]	10 [100]
TQA*	-	-	-
TQA* (Aer)	10 [70]	10 [100]	10 [100]
PP			
Fixed Angle*	10 [70]	10 [100]	10 [100]
Fixed Angle†	10 [70]	10 [100]	10 [100]
Fourier*	10 [70]	10 [100]	10 [100]
Interp.	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	10 [70]	10 [100]	10 [100]
TQA*	10 [70]	10 [100]	10 [100]
SV			
Fixed Angle*	-	-	10 [20]
Fixed Angle†	-	-	10 [20]
Fourier*	-	-	10 [20]
Interp.	-	-	10 [20]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	10 [20]
TQA*	-	-	10 [20]

Table 40: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	11995.1 $\pm$ 3162.9	22494.0 $\pm$ 518.0	17239.6 $\pm$ 49.8
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-12244.9 $\pm$ 3869.0	8579.6 $\pm$ 19.7	14350.8 $\pm$ 279.9
Fourier (Aer)	65073.4 $\pm$ 2065.2	23889.9 $\pm$ 331.7	17227.5 $\pm$ 109.2
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	30289.6 $\pm$ 18260.2	22168.3 $\pm$ 982.7	17355.5 $\pm$ 48.4
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	45337.4 $\pm$ 4065.1	21463.2 $\pm$ 18.7	16587.7 $\pm$ 14.7
TQA*	-	-	-
TQA* (Aer)	52894.8 $\pm$ 6299.2	23803.7 $\pm$ 314.5	17309.4 $\pm$ 25.9
PP			
Fixed Angle*	6774.6 $\pm$ 6735.8	20203.6 $\pm$ 0.0	16592.8 $\pm$ 16.9
Fixed Angle [†]	-2308.7 $\pm$ 383.9	5827.3 $\pm$ 0.0	6629.2 $\pm$ 24.3
Fourier*	73002.8 $\pm$ 3089.2	25757.2 $\pm$ 0.0	17619.5 $\pm$ 15.3
Interp.	27879.7 $\pm$ 27264.2	23790.6 $\pm$ 0.0	17580.0 $\pm$ 17.3
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	18316.7 $\pm$ 5870.9	21509.8 $\pm$ 0.0	16273.7 $\pm$ 4.3
TQA*	57385.8 $\pm$ 12377.5	24749.9 $\pm$ 0.0	17512.7 $\pm$ 19.7
SV			
Fixed Angle*	-	-	3325.0 $\pm$ 11.3
Fixed Angle [†]	-	-	1332.7 $\pm$ 20.0
Fourier*	-	-	3466.2 $\pm$ 8.9
Interp.	-	-	3444.1 $\pm$ 5.1
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	3259.8 $\pm$ 3.1
TQA*	-	-	3479.2 $\pm$ 1.4

Table 41: **Avg. Energy $\pm$ standard deviation for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	L2F	RR	ER	L2F	RR
MPS						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle* (Aer)	11995.1 $\pm$ 3162.9	22494.0 $\pm$ 518.0	17239.6 $\pm$ 49.8	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-12244.9 $\pm$ 3869.0	8579.6 $\pm$ 19.7	14350.8 $\pm$ 279.9	10 [70]	10 [100]	10 [100]
Fourier (Aer)	65073.4 $\pm$ 2065.2	23889.9 $\pm$ 331.7	17227.5 $\pm$ 109.2	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-	-	-	-
Interp.	-	-	-	-	-	-
Interp. (Aer)	30289.6 $\pm$ 18260.2	22168.3 $\pm$ 982.7	17355.5 $\pm$ 48.4	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA (Aer)	45337.4 $\pm$ 4065.1	21463.2 $\pm$ 18.7	16587.7 $\pm$ 14.7	10 [70]	10 [100]	10 [100]
TQA*	-	-	-	-	-	-
TQA* (Aer)	52894.8 $\pm$ 6299.2	23803.7 $\pm$ 314.5	17309.4 $\pm$ 25.9	10 [70]	10 [100]	10 [100]
PP						
Fixed Angle*	6774.6 $\pm$ 6735.8	20203.6 $\pm$ 0.0	16592.8 $\pm$ 16.9	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-2308.7 $\pm$ 383.9	5827.3 $\pm$ 0.0	6629.2 $\pm$ 24.3	10 [70]	10 [100]	10 [100]
Fourier*	73002.8 $\pm$ 3089.2	25757.2 $\pm$ 0.0	17619.5 $\pm$ 15.3	10 [70]	10 [100]	10 [100]
Interp.	27879.7 $\pm$ 27264.2	23790.6 $\pm$ 0.0	17580.0 $\pm$ 17.3	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	18316.7 $\pm$ 5870.9	21509.8 $\pm$ 0.0	16273.7 $\pm$ 4.3	10 [70]	10 [100]	10 [100]
TQA*	57385.8 $\pm$ 12377.5	24749.9 $\pm$ 0.0	17512.7 $\pm$ 19.7	10 [70]	10 [100]	10 [100]
SV						
Fixed Angle*	-	-	3325.0 $\pm$ 11.3	-	-	10 [20]
Fixed Angle [†]	-	-	1332.7 $\pm$ 20.0	-	-	10 [20]
Fourier*	-	-	3466.2 $\pm$ 8.9	-	-	10 [20]
Interp.	-	-	3444.1 $\pm$ 5.1	-	-	10 [20]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	3259.8 $\pm$ 3.1	-	-	10 [20]
TQA*	-	-	3479.2 $\pm$ 1.4	-	-	10 [20]

Table 42: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

15 Tables for depth $P = 5$

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	-	-	-
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-	-	-
Fourier (Aer)	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	-	-	-
TQA*	-	-	-
TQA* (Aer)	-	-	-
PP			
Fixed Angle*	-	-	-
Fixed Angle [†]	-	-	-
Fourier*	10 [70]	10 [100]	10 [100]
Interp.	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fixed Angle*	-	-	10 [20]
Fixed Angle [†]	-	-	10 [20]
Fourier*	-	-	10 [20]
Interp.	-	-	10 [20]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	10 [20]
TQA*	-	-	10 [20]

Table 43: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	-	-	-
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-	-	-
Fourier (Aer)	65502.3 ± 2332.4	24101.5 ± 259.9	17355.1 ± 70.2
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	27765.9 ± 13747.4	22601.5 ± 712.6	17424.0 ± 38.1
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	-	-	-
TQA*	-	-	-
TQA* (Aer)	-	-	-
PP			
Fixed Angle*	-	-	-
Fixed Angle [†]	-	-	-
Fourier*	79758.6 ± 2424.4	26076.0 ± 0.0	17773.9 ± 20.7
Interp.	35592.9 ± 25747.1	19595.6 ± 0.0	17756.7 ± 27.4
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fixed Angle*	-	-	3184.5 ± 176.6
Fixed Angle [†]	-	-	1129.2 ± 9.2
Fourier*	-	-	3478.5 ± 5.5
Interp.	-	-	3462.5 ± 3.6
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	3297.7 ± 5.7
TQA*	-	-	3479.0 ± 1.9

Table 44: **Avg. Energy ± standard deviation for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	L2F	RR	ER	L2F	RR
MPS						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle* (Aer)	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-	-	-	-	-	-
Fourier (Aer)	65502.3 $\pm$ 2332.4	24101.5 $\pm$ 259.9	17355.1 $\pm$ 70.2	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-	-	-	-
Interp.	-	-	-	-	-	-
Interp. (Aer)	27765.9 $\pm$ 13747.4	22601.5 $\pm$ 712.6	17424.0 $\pm$ 38.1	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
TQA* (Aer)	-	-	-	-	-	-
PP						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-
Fourier*	79758.6 $\pm$ 2424.4	26076.0 $\pm$ 0.0	17773.9 $\pm$ 20.7	10 [70]	10 [100]	10 [100]
Interp.	35592.9 $\pm$ 25747.1	19595.6 $\pm$ 0.0	17756.7 $\pm$ 27.4	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
SV						
Fixed Angle*	-	-	3184.5 $\pm$ 176.6	-	-	10 [20]
Fixed Angle [†]	-	-	1129.2 $\pm$ 9.2	-	-	10 [20]
Fourier*	-	-	3478.5 $\pm$ 5.5	-	-	10 [20]
Interp.	-	-	3462.5 $\pm$ 3.6	-	-	10 [20]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	3297.7 $\pm$ 5.7	-	-	10 [20]
TQA*	-	-	3479.0 $\pm$ 1.9	-	-	10 [20]

Table 45: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

16 Tables for depth $P = 6$

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	10 [70]	10 [100]	10 [100]
Fourier (Aer)	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	10 [70]	10 [100]	10 [100]
TQA*	-	-	-
TQA* (Aer)	10 [70]	10 [100]	10 [100]
PP			
Fixed Angle*	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	10 [70]	10 [100]	10 [100]
Fourier*	10 [70]	10 [100]	10 [100]
Interp.	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	10 [70]	10 [100]	10 [100]
TQA*	10 [70]	10 [100]	10 [100]
SV			
Fixed Angle*	-	-	10 [20]
Fixed Angle [†]	-	-	10 [20]
Fourier*	-	-	10 [20]
Interp.	-	-	10 [20]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	5 [20]	-	10 [20]
TQA*	-	-	10 [20]

Table 46: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	L2F	RR
MPS			
Fixed Angle*	-	-	-
Fixed Angle* (Aer)	16048.7 $\pm$ 7754.3	22541.1 $\pm$ 385.0	17173.4 $\pm$ 50.6
Fixed Angle [†]	-	-	-
Fixed Angle [†] (Aer)	-18046.0 $\pm$ 3622.7	7168.5 $\pm$ 29.9	16194.0 $\pm$ 92.0
Fourier (Aer)	65882.6 $\pm$ 2605.2	24117.0 $\pm$ 532.0	17398.2 $\pm$ 62.9
Fourier*	-	-	-
Interp.	-	-	-
Interp. (Aer)	41477.0 $\pm$ 10764.6	22849.9 $\pm$ 266.4	17427.2 $\pm$ 68.3
Linear Ramp*	-	-	-
Linear Ramp* (Aer)	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA (Aer)	12617.7 $\pm$ 2917.7	21160.4 $\pm$ 21.2	16771.7 $\pm$ 14.6
TQA*	-	-	-
TQA* (Aer)	32061.9 $\pm$ 11764.8	24234.3 $\pm$ 142.4	17309.3 $\pm$ 40.8
PP			
Fixed Angle*	13159.7 $\pm$ 13436.7	23810.2 $\pm$ 0.0	17248.7 $\pm$ 57.6
Fixed Angle [†]	-1659.8 $\pm$ 263.1	1593.1 $\pm$ 0.0	10487.2 $\pm$ 18.4
Fourier*	78103.2 $\pm$ 7976.1	25853.4 $\pm$ 0.0	17940.8 $\pm$ 30.8
Interp.	55315.5 $\pm$ 32067.7	25263.7 $\pm$ 0.0	17786.4 $\pm$ 18.8
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	3285.8 $\pm$ 753.9	18897.2 $\pm$ 0.0	16844.9 $\pm$ 5.5
TQA*	19827.1 $\pm$ 3918.3	25357.7 $\pm$ 0.0	17627.0 $\pm$ 81.7
SV			
Fixed Angle*	-	-	3390.5 $\pm$ 39.3
Fixed Angle [†]	-	-	2074.4 $\pm$ 28.2
Fourier*	-	-	3468.6 $\pm$ 13.6
Interp.	-	-	3460.2 $\pm$ 3.6
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	1715.3 $\pm$ 1036.7	-	3304.4 $\pm$ 1.8
TQA*	-	-	3481.8 $\pm$ 3.1

Table 47: **Avg. Energy $\pm$ standard deviation for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	L2F	RR	ER	L2F	RR
MPS						
Fixed Angle*	-	-	-	-	-	-
Fixed Angle* (Aer)	16048.7 $\pm$ 7754.3	22541.1 $\pm$ 385.0	17173.4 $\pm$ 50.6	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-	-	-	-	-	-
Fixed Angle [†] (Aer)	-18046.0 $\pm$ 3622.7	7168.5 $\pm$ 29.9	16194.0 $\pm$ 92.0	10 [70]	10 [100]	10 [100]
Fourier (Aer)	65882.6 $\pm$ 2605.2	24117.0 $\pm$ 532.0	17398.2 $\pm$ 62.9	10 [70]	10 [100]	10 [100]
Fourier*	-	-	-	-	-	-
Interp.	-	-	-	-	-	-
Interp. (Aer)	41477.0 $\pm$ 10764.6	22849.9 $\pm$ 266.4	17427.2 $\pm$ 68.3	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Linear Ramp* (Aer)	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA (Aer)	12617.7 $\pm$ 2917.7	21160.4 $\pm$ 21.2	16771.7 $\pm$ 14.6	10 [70]	10 [100]	10 [100]
TQA*	-	-	-	-	-	-
TQA* (Aer)	32061.9 $\pm$ 11764.8	24234.3 $\pm$ 142.4	17309.3 $\pm$ 40.8	10 [70]	10 [100]	10 [100]
PP						
Fixed Angle*	13159.7 $\pm$ 13436.7	23810.2 $\pm$ 0.0	17248.7 $\pm$ 57.6	10 [70]	10 [100]	10 [100]
Fixed Angle [†]	-1659.8 $\pm$ 263.1	1593.1 $\pm$ 0.0	10487.2 $\pm$ 18.4	10 [70]	10 [100]	10 [100]
Fourier*	78103.2 $\pm$ 7976.1	25853.4 $\pm$ 0.0	17940.8 $\pm$ 30.8	10 [70]	10 [100]	10 [100]
Interp.	55315.5 $\pm$ 32067.7	25263.7 $\pm$ 0.0	17786.4 $\pm$ 18.8	10 [70]	10 [100]	10 [100]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	3285.8 $\pm$ 753.9	18897.2 $\pm$ 0.0	16844.9 $\pm$ 5.5	10 [70]	10 [100]	10 [100]
TQA*	19827.1 $\pm$ 3918.3	25357.7 $\pm$ 0.0	17627.0 $\pm$ 81.7	10 [70]	10 [100]	10 [100]
SV						
Fixed Angle*	-	-	3390.5 $\pm$ 39.3	-	-	10 [20]
Fixed Angle [†]	-	-	2074.4 $\pm$ 28.2	-	-	10 [20]
Fourier*	-	-	3468.6 $\pm$ 13.6	-	-	10 [20]
Interp.	-	-	3460.2 $\pm$ 3.6	-	-	10 [20]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	1715.3 $\pm$ 1036.7	-	3304.4 $\pm$ 1.8	5 [20]	-	10 [20]
TQA*	-	-	3481.8 $\pm$ 3.1	-	-	10 [20]

Table 48: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

17 Tables for depth $P = 7$

	RR
MPS	
Fixed Angle*	-
Fixed Angle* (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier*	-
Interp.	-
Interp. (Aer)	-
Linear Ramp*	-
Linear Ramp* (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA*	-
TQA* (Aer)	-
PP	
Fixed Angle*	-
Fixed Angle [†]	-
Fourier*	-
Interp.	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fixed Angle*	10 [20]
Fixed Angle [†]	10 [20]
Fourier*	10 [20]
Interp.	10 [20]
Linear Ramp*	-
Recursive TS	9 [20]
TQA	10 [20]
TQA*	10 [20]

Table 49: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fixed Angle [*]	-
Fixed Angle [*] (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier [*]	-
Interp.	-
Interp. (Aer)	-
Linear Ramp [*]	-
Linear Ramp [*] (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA [*]	-
TQA [*] (Aer)	-
PP	
Fixed Angle [*]	-
Fixed Angle [†]	-
Fourier [*]	-
Interp.	-
Linear Ramp [*]	-
Recursive TS	-
TQA	-
TQA [*]	-
SV	
Fixed Angle [*]	3237.2 ± 47.0
Fixed Angle [†]	1666.7 ± 10.3
Fourier [*]	3459.9 ± 11.8
Interp.	3462.4 ± 2.2
Linear Ramp [*]	-
Recursive TS	3466.4 ± 13.2
TQA	3256.5 ± 3.3
TQA [*]	3492.1 ± 1.1

Table 50: **Avg. Energy ± standard deviation for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fixed Angle [*]	-	-
Fixed Angle [*] (Aer)	-	-
Fixed Angle [†]	-	-
Fixed Angle [†] (Aer)	-	-
Fourier (Aer)	-	-
Fourier [*]	-	-
Interp.	-	-
Interp. (Aer)	-	-
Linear Ramp [*]	-	-
Linear Ramp [*] (Aer)	-	-
Recursive TS	-	-
TQA	-	-
TQA (Aer)	-	-
TQA [*]	-	-
TQA [*] (Aer)	-	-
PP		
Fixed Angle [*]	-	-
Fixed Angle [†]	-	-
Fourier [*]	-	-
Interp.	-	-
Linear Ramp [*]	-	-
Recursive TS	-	-
TQA	-	-
TQA [*]	-	-
SV		
Fixed Angle [*]	3237.2 $\pm$ 47.0	10 [20]
Fixed Angle [†]	1666.7 $\pm$ 10.3	10 [20]
Fourier [*]	3459.9 $\pm$ 11.8	10 [20]
Interp.	3462.4 $\pm$ 2.2	10 [20]
Linear Ramp [*]	-	-
Recursive TS	3466.4 $\pm$ 13.2	9 [20]
TQA	3256.5 $\pm$ 3.3	10 [20]
TQA [*]	3492.1 $\pm$ 1.1	10 [20]

Table 51: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

18 Tables for depth $P = 8$

	RR
MPS	
Fixed Angle*	-
Fixed Angle* (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier*	-
Interp.	-
Interp. (Aer)	-
Linear Ramp*	-
Linear Ramp* (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA*	-
TQA* (Aer)	-
PP	
Fixed Angle*	-
Fixed Angle [†]	-
Fourier*	-
Interp.	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fixed Angle*	10 [20]
Fixed Angle [†]	10 [20]
Fourier*	10 [20]
Interp.	10 [20]
Linear Ramp*	-
Recursive TS	-
TQA	10 [20]
TQA*	10 [20]

Table 52: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fixed Angle*	-
Fixed Angle* (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier*	-
Interp.	-
Interp. (Aer)	-
Linear Ramp*	-
Linear Ramp* (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA*	-
TQA* (Aer)	-
PP	
Fixed Angle*	-
Fixed Angle [†]	-
Fourier*	-
Interp.	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fixed Angle*	3253.9 ± 440.8
Fixed Angle [†]	2042.3 ± 32.5
Fourier*	3447.5 ± 18.4
Interp.	3447.1 ± 6.4
Linear Ramp*	-
Recursive TS	-
TQA	3331.9 ± 3.3
TQA*	3461.8 ± 7.9

Table 53: **Avg. Energy ± standard deviation for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fixed Angle*	-	-
Fixed Angle* (Aer)	-	-
Fixed Angle [†]	-	-
Fixed Angle [†] (Aer)	-	-
Fourier (Aer)	-	-
Fourier*	-	-
Interp.	-	-
Interp. (Aer)	-	-
Linear Ramp*	-	-
Linear Ramp* (Aer)	-	-
Recursive TS	-	-
TQA	-	-
TQA (Aer)	-	-
TQA*	-	-
TQA* (Aer)	-	-
PP		
Fixed Angle*	-	-
Fixed Angle [†]	-	-
Fourier*	-	-
Interp.	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fixed Angle*	3253.9 ± 440.8	10 [20]
Fixed Angle [†]	2042.3 ± 32.5	10 [20]
Fourier*	3447.5 ± 18.4	10 [20]
Interp.	3447.1 ± 6.4	10 [20]
Linear Ramp*	-	-
Recursive TS	-	-
TQA	3331.9 ± 3.3	10 [20]
TQA*	3461.8 ± 7.9	10 [20]

Table 54: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

19 Tables for depth $P = 9$

	RR
MPS	
Fixed Angle*	-
Fixed Angle* (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier*	-
Interp.	-
Interp. (Aer)	-
Linear Ramp*	-
Linear Ramp* (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA*	-
TQA* (Aer)	-
PP	
Fixed Angle*	-
Fixed Angle [†]	-
Fourier*	-
Interp.	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fixed Angle*	10 [20]
Fixed Angle [†]	10 [20]
Fourier*	10 [20]
Interp.	10 [20]
Linear Ramp*	-
Recursive TS	-
TQA	10 [20]
TQA*	10 [20]

Table 55: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fixed Angle*	-
Fixed Angle* (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier*	-
Interp.	-
Interp. (Aer)	-
Linear Ramp*	-
Linear Ramp* (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA*	-
TQA* (Aer)	-
PP	
Fixed Angle*	-
Fixed Angle [†]	-
Fourier*	-
Interp.	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fixed Angle*	3032.3 $\pm$ 655.3
Fixed Angle [†]	1161.7 $\pm$ 22.8
Fourier*	3442.1 $\pm$ 19.5
Interp.	3441.1 $\pm$ 8.1
Linear Ramp*	-
Recursive TS	-
TQA	3304.2 $\pm$ 4.3
TQA*	3465.1 $\pm$ 8.1

Table 56: **Avg. Energy $\pm$ standard deviation for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fixed Angle [*]	-	-
Fixed Angle [*] (Aer)	-	-
Fixed Angle [†]	-	-
Fixed Angle [†] (Aer)	-	-
Fourier (Aer)	-	-
Fourier [*]	-	-
Interp.	-	-
Interp. (Aer)	-	-
Linear Ramp [*]	-	-
Linear Ramp [*] (Aer)	-	-
Recursive TS	-	-
TQA	-	-
TQA (Aer)	-	-
TQA [*]	-	-
TQA [*] (Aer)	-	-
PP		
Fixed Angle [*]	-	-
Fixed Angle [†]	-	-
Fourier [*]	-	-
Interp.	-	-
Linear Ramp [*]	-	-
Recursive TS	-	-
TQA	-	-
TQA [*]	-	-
SV		
Fixed Angle [*]	3032.3 $\pm$ 655.3	10 [20]
Fixed Angle [†]	1161.7 $\pm$ 22.8	10 [20]
Fourier [*]	3442.1 $\pm$ 19.5	10 [20]
Interp.	3441.1 $\pm$ 8.1	10 [20]
Linear Ramp [*]	-	-
Recursive TS	-	-
TQA	3304.2 $\pm$ 4.3	10 [20]
TQA [*]	3465.1 $\pm$ 8.1	10 [20]

Table 57: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

20 Tables for depth $P = 10$

	RR
MPS	
Fixed Angle*	-
Fixed Angle* (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier*	-
Interp.	-
Interp. (Aer)	-
Linear Ramp*	-
Linear Ramp* (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA*	-
TQA* (Aer)	-
PP	
Fixed Angle*	-
Fixed Angle [†]	-
Fourier*	-
Interp.	-
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fixed Angle*	10 [20]
Fixed Angle [†]	10 [20]
Fourier*	10 [20]
Interp.	10 [20]
Linear Ramp*	-
Recursive TS	1 [20]
TQA	10 [20]
TQA*	10 [20]

Table 58: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fixed Angle [*]	-
Fixed Angle [*] (Aer)	-
Fixed Angle [†]	-
Fixed Angle [†] (Aer)	-
Fourier (Aer)	-
Fourier [*]	-
Interp.	-
Interp. (Aer)	-
Linear Ramp [*]	-
Linear Ramp [*] (Aer)	-
Recursive TS	-
TQA	-
TQA (Aer)	-
TQA [*]	-
TQA [*] (Aer)	-
PP	
Fixed Angle [*]	-
Fixed Angle [†]	-
Fourier [*]	-
Interp.	-
Linear Ramp [*]	-
Recursive TS	-
TQA	-
TQA [*]	-
SV	
Fixed Angle [*]	3382.9 ± 19.6
Fixed Angle [†]	2014.8 ± 26.6
Fourier [*]	3414.5 ± 35.6
Interp.	3418.3 ± 11.3
Linear Ramp [*]	-
Recursive TS	3447.7
TQA	3277.5 ± 2.6
TQA [*]	3463.6 ± 9.6

Table 59: **Avg. Energy ± standard deviation for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fixed Angle [*]	-	-
Fixed Angle [*] (Aer)	-	-
Fixed Angle [†]	-	-
Fixed Angle [†] (Aer)	-	-
Fourier (Aer)	-	-
Fourier [*]	-	-
Interp.	-	-
Interp. (Aer)	-	-
Linear Ramp [*]	-	-
Linear Ramp [*] (Aer)	-	-
Recursive TS	-	-
TQA	-	-
TQA (Aer)	-	-
TQA [*]	-	-
TQA [*] (Aer)	-	-
PP		
Fixed Angle [*]	-	-
Fixed Angle [†]	-	-
Fourier [*]	-	-
Interp.	-	-
Linear Ramp [*]	-	-
Recursive TS	-	-
TQA	-	-
TQA [*]	-	-
SV		
Fixed Angle [*]	3382.9 ± 19.6	10 [20]
Fixed Angle [†]	2014.8 ± 26.6	10 [20]
Fourier [*]	3414.5 ± 35.6	10 [20]
Interp.	3418.3 ± 11.3	10 [20]
Linear Ramp [*]	-	-
Recursive TS	3447.7	1 [20]
TQA	3277.5 ± 2.6	10 [20]
TQA [*]	3463.6 ± 9.6	10 [20]

Table 60: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.